Measuring the Gap: Semantic Alignment Between AI-for-Sustainability Research and SDG Policy Frameworks

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Abstract

The Sustainable Development Goals (SDGs) are the dominant international development framework. Shared SDG labels do not necessarily imply shared semantic framing between research and policy. Existing bibliometric studies map AI research to SDGs using keyword or classification methods (Armitage et al., 2020; Kashnitsky et al., 2024), but rarely compare the semantic content of research and policy discourse. This study uses Sentence-BERT embeddings and a supervised logistic regression classifier (test macro-F1 = 0.817) to map semantic distance between 3,105,144 AI-for-sustainability abstracts and 40,597 segments from 6,367 institutional SDG policy sources. The analysis separates coverage gaps (which SDGs each corpus emphasises) from within-SDG semantic gaps (how far texts assigned to the same SDG sit apart). Because the corpora differ in register (Biber and Conrad, 2009), these gaps are observed textual-semantic divergence, not substantive disagreement.

The central finding is a dissociation: the coverage gap and the within-SDG semantic gap are independent dimensions of divergence, not linked effects. The primary hypothesis (H1a) tests whether the coverage gap predicts semantic divergence: the association is near-zero in the canonical specification ($r = -0.000$, $p = 0.999$). Research coverage alone likewise does not predict semantic divergence (H1c: $r = 0.123$, $p = 0.637$; $n = 17$). No hypothesis survives robustness checks across alternative assignment methods, policy source families, or segment caps. Together, these null results show that coverage and framing should be measured as separate axes of divergence. The study reports three supporting patterns. First, policy discourse concentrates on SDGs 16 (Peace and Strong Institutions, 17.5%), 13 (Climate Action, 11.3%), and 17 (Partnerships, 11.4%), together 40.2% of policy coverage. Second, research concentrates on SDGs 3 (Good Health, 26.0%), 9 (Industry and Infrastructure, 18.5%), and 4 (Quality Education, artefact-affected, 14.6%). Third, the largest within-SDG semantic gaps occur in SDGs 13 (0.481), 3 (0.470), 11 (0.464), 1 (0.447), and 7 (0.418).

These findings do not prove substantive research-policy misalignment. They show that shared SDG labels are insufficient evidence of shared semantic framing, and that embedding-based distance is a transparent diagnostic map of proximity and divergence. The results need caution because of register effects, policy source-family heterogeneity, OpenAlex retrieval boundaries, and an SDG 4 assignment that is artefact-affected rather than a clean education finding.

1 Introduction

Research-policy alignment is often inferred from whether scientific and policy texts address the same topics. Yet shared topical coverage does not establish that both sides frame those topics in comparable ways: a paper and a policy document can both invoke SDG 13

(Climate Action) or SDG 3 (Good Health) while emphasising different problems, actors, and interventions. High-level thematic agreement can therefore mask deeper divergence in how each discourse community approaches the same goal.

Existing methods for mapping research to the Sustainable Development Goals (SDGs) operate mainly at the level of lexical or bibliometric correspondence: keyword dictionaries, Boolean queries, or citation and mention counts (Armitage et al., 2020; Kashnitsky et al., 2024). They measure whether a text is *about* a given SDG, not how it frames it. That interchangeability is what this paper questions.

This paper proposes that research-policy alignment has at least two separable dimensions: coverage divergence (which SDGs each corpus prioritises) and semantic framing divergence (how differently they discuss the same SDG). Its contribution is a measurement framework. I advance AI–SDG mapping from Level 1 (lexical correspondence) to Level 2 (observed textual-semantic proximity in embedding space; Section 2.2), treating semantic proximity as a register-sensitive proxy for framing, not evidence of substantive alignment. I operationalise both dimensions across all 17 SDGs with frozen Sentence-BERT embeddings (Reimers & Gurevych, 2019) and a supervised logistic-regression classifier, and stress-test the instrument against its principal measurement choices.

The central finding is a dissociation: coverage and framing are separate dimensions of divergence, not linked effects. This matters because SDG tagging is now institutionalised: OpenAlex labels works, funders count SDG outputs, and journals print SDG badges. If shared labels are structurally disconnected from shared framing, that threatens how the SDG-monitoring ecosystem reads its own signals. The divergence that survives robustness checks concentrates in a small set of goals: SDG 13 is consistently the most divergent, SDG 15 the least, and SDG 17 the most sensitive to how divergence is measured. I report this pattern with its scope conditions.

The central research question is: *To what extent do academic AI-for-sustainability research and international institutional SDG policy discourse differ in SDG coverage and semantic framing across the 17 SDGs, and how are these two dimensions related?* Section 2 reviews the literature and introduces the three-level framework; Section 3 describes the corpora, measurement instrument, and gap procedures; Sections 4 and 5 report and interpret the findings.

2 Literature Review

2.1 AI and the SDGs: What Bibliometrics Shows

This section reviews past studies on AI and the SDGs. Most ask which goals draw the most research attention.

The best-known work, such as Vinuesa et al. (2020)'s assessment of AI's effects on the 169 SDG targets, rests on expert views or counts rather than on the texts themselves.

Broader SDG scientometric work has mapped the growth, institutional distribution, thematic structure, and interlinkages of MDG/SDG-related research across methods and time windows (Bautista-Puig et al., 2021; Cowls et al., 2021; Nedungadi et al., 2024; Singh et al., 2024). Bautista-Puig et al. (2021) chart the general SDG research landscape across higher-education and research institutions from 2000 to 2017 using a citation-based corpus of roughly 25,000 publications, finding SDG 3 (Good Health) by far the most addressed goal and SDG 7 and SDG 12 among the least. The AI-specific inventories converge on SDG 3 as dominant: Singh et al. (2024) map 20,511 AI publications (2001–2020) and rank SDG 3 and SDG 7 as the most applications-rich goals with SDG 17 the sparsest, while Cowls et al. (2021) survey 108 AI-for-social-good projects and report SDG 3 as the leader and SDGs 5, 16, and 17 each addressed by fewer than five. Nedungadi et al. (2024) apply BERTopic modelling to 1,288 big-data and AI publications (2013–2023) and likewise find SDG 3 most prominent, with SDG 9 and SDG 7 next. Across these studies, SDG 3 is the one robustly dominant goal, while the least-covered goals differ between the AI-specific inventories and the broader bibliometric picture — but none of them measures within-SDG semantic framing.

The methodological concern raised by Armitage et al. (2020) is important: SDG publication mapping depends on interpretive choices about what counts as SDG relevance, how target concepts are translated into queries or labels, and which bibliographic database is used. The embedding-based approach used here reduces some vocabulary-level brittleness of keyword dictionaries, but it remains a distributional measurement strategy whose results are conditional on corpus construction, centroid provenance, and model choice. Later benchmarking work reaches the same caution from a broader comparison of Boolean, machine-learning, and hybrid SDG-mapping systems: Kashnitsky et al. (2024) find that no single approach performs best across all validation datasets, and that no single "golden" SDG validation set exists.

Not only are SDG mapping systems methodologically contingent; the SDGs themselves are interdependent by design. Nilsson et al. (2016) observe that "the goals depend on each other" but that no operational framework specifies how, warning that treating targets in isolation "risk[s] perverse outcomes." Single-SDG assignment is therefore a simplifying assumption, not a reflection of how goals operate in practice.

One SDG breaks the interdependence pattern. Le Blanc (2015), in a network analysis of the proposed SDGs, treats SDG 17 (Partnerships for the Goals) as the dedicated goal for cross-cutting means of implementation — finance, technology transfer, capacity-building, and partnerships — and excludes it from cross-goal network mapping because its targets are implementation infrastructure rather than substantive domains: a distinction unique among the 17 goals. The Griggs et al. (2017) guide likewise describes SDG 17 as the main enabler of the whole framework. Unlike every other SDG, whose discourse centres on a shared substantive topic, SDG 17 policy discourse concerns coordinating and resourcing implementation itself — a conceptual space that technical AI research abstracts are not built to occupy.

Past work shows which SDGs attract AI and sustainability research, and how much that mapping depends on method and corpus. Whether research and policy texts for the same goals actually align is the subject of the next section.

2.2 A Three-Level Framework of Research-Policy Alignment

This paper proposes that research-policy alignment be understood as a three-level construct rather than a binary property of shared labels:

Level 1 — lexical or bibliometric correspondence. The dominant mode of existing SDG-mapping work, assigning texts to goals through keyword dictionaries, Boolean queries, or citation and mention counts (Armitage et al., 2020; Kashnitsky et al., 2024). Necessary but weak: shared vocabulary does not guarantee shared meaning (Bornmann et al., 2016), and keyword methods are brittle to vocabulary and database choices (Armitage et al., 2020).

Level 2 — observed textual-semantic proximity in embedding space. Sentence embeddings locate texts in a shared semantic space where proximity reflects distributional patterns of co-occurrence (Harris, 1954; Reimers & Gurevych, 2019), and encoder-based SDG association has been benchmarked as a competitive baseline (Gjorgjevikj et al., 2025). This level detects divergence that keyword methods miss, while remaining a proxy rather than a definitive conclusion: embedding distance mixes modality, abstraction, institutional stance, and theme (Biber, 1988), and the research-policy contrast is primarily a register contrast (Biber & Conrad, 2009). The present study is positioned at this level.

Level 3 — substantive comparison of implied problems, actors, and interventions. The ultimate object of alignment is whether the two discourses address the same problems, involve the same actors, and propose the same interventions (Cash et al., 2003; Guston, 2001; Haas, 1992). Because this requires interpreting causal beliefs, legitimacy, and institutional mechanism (Caplan, 1979; Haas, 1992), it lies beyond unsupervised text analysis and is left to qualitative follow-up.

Advancing from Level 1 to Level 2 is the study's concrete methodological contribution; Level 3 names the boundary it deliberately does not cross. Figure 1 illustrates this decomposition.

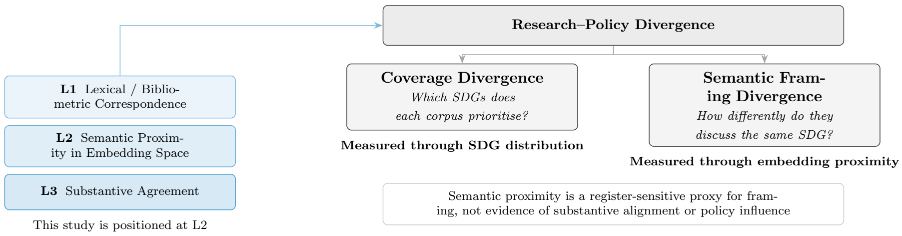

Figure 1. Conceptual framework: research–policy divergence separated into coverage and semantic framing dimensions, positioned within the three-level alignment construct.

Figure 3 summarises the complete data pipeline (Section 3.11).

2.3 Research-Policy Alignment in AI and Sustainability

This section asks whether AI research and sustainability policy line up, and why past work says they often do not.

A growing literature documents structural tensions between what AI research produces and what sustainability policy requires. Strauss et al. (2025) show that corporate AI research concentrates on pre-deployment technical problems, creating a gap between research priorities and the real-world deployment impacts that governance and policy must address. Sioumalas-Christodoulou and Tympas (2025) find that global AI metrics focus on technical and economic performance while policy discourse emphasises ethical and social dimensions aligned with SDG-related concerns. Toney-Wails et al. (2024), in the trustworthy-AI governance literature, provide a close methodological precedent, showing that policy and research communities can use shared trustworthy-AI terminology while differing in term salience, definitions, and focal concerns.

McCarthy et al. (2026) report a parallel finding in conservation science, where a multi-target SciBERT classifier applied to Ross Sea MPA research reveals that 59% of policy-prioritised research topics appear in fewer than 15 papers, demonstrating that research-policy coverage gaps are observable across domains.

Adjacent scientometric work has also used policy documents to measure research-policy connection, but at the level of mentions rather than semantic framing. Bornmann et al. (2016) examine climate-change publications in Altmetric-tracked policy documents and find that only 1.2% of 191,276 publications had at least one policy mention. Their study is useful here as a boundary marker: policy-document mentions can indicate one narrow form of policy uptake, but they do not show how research is discussed, interpreted, or framed inside policy discourse. The present study therefore addresses a different layer of the research-policy relationship: not whether policy documents cite particular papers, but whether research and policy-facing texts assigned to the same SDG occupy nearby regions in semantic space.

A related research-evaluation literature treats the research-policy relationship through the lens of societal impact measurement. Bornmann (2017) argues that measuring societal impact is harder than measuring scientific impact because the target of impact must first be specified: research may affect policymakers, business, culture, public health, or other social domains. Policy documents are therefore useful as one possible target-oriented source for measuring impact on policymaking, but this is a different task from semantic framing comparison. The present study does not ask whether research has achieved policy impact; it asks whether research and policy-facing texts occupy similar semantic regions when organised through shared SDG labels.

These findings are consistent with theoretical accounts of research-policy misalignment. Heink et al. (2015) identify credibility, relevance, and legitimacy as determinants of whether scientific knowledge informs policy; topical relevance alone is insufficient — relevance also depends on whether research results can shape actual decisions. In sustainability science, Cash et al. (2003) make a closely related argument: knowledge systems are more likely to support action when they produce information that is simultaneously salient, credible, and legitimate, and when they manage the boundary between knowledge and action through communication, translation, and mediation. The epistemic communities framework (Haas, 1992) provides a theoretical reason why shared framing matters for policy coordination under uncertainty: expert communities can influence policy by articulating cause–effect relationships, framing issues for collective debate, proposing policies, and identifying salient points for negotiation. However, Haas's concept is broader than semantic proximity alone, involving shared normative beliefs, causal beliefs, validity criteria, and a common policy enterprise. The present study therefore does not operationalise an epistemic community directly; it measures the weaker condition of observed semantic proximity between research and policy-facing discourse.

Cross (2013) strengthens this caution by arguing that epistemic communities should not be treated as simply present or absent: their influence varies with internal cohesion, professionalism, access to decision-makers, competition with other actors, and broader political context. This means that semantic proximity can at most indicate a possible precondition for knowledge transfer, not the existence, strength, or influence of an epistemic community.

The SDG framework is itself a politically negotiated instrument with documented limitations. Fukuda-Parr (2016) argues that global goals can have "distorting effects, redefining the meaning of development, and shaping policy by creating incentives to over-focus on target achievement." Bexell and Jönsson (2017) find that SDG summit documents remain "state-centric with great room for state sovereignty," avoid assigning causal responsibility for structural problems, and rely on "weak and unsystematic accountability."

A more fundamental tension exists within the framework itself. Hickel (2019) demonstrates that "Goal 8 violates the sustainability objectives of the SDGs" — 3% annual global growth is empirically incompatible with the resource-use and emissions targets in Goals 6, 12, 13, 14, and 15. The SDGs "lack theoretical grounding" and embody a contradiction between growth and ecological sustainability.

Heilinger et al. (2024) warns that the "sustainable AI" label can be used for greenwashing where technologies are framed as contributing to the SDGs without sufficient basis. Heras-Saizarbitoria et al. (2022), analysing 1,370 sustainability reports, find that most organisations' SDG engagement is superficial — "the SDGs only serve to add colour and fancy icons to reports" — suggesting that SDG-washing is a measurable empirical pattern. This supports the caution that reported alignment should not be conflated with substantive commitment.

Measuring semantic alignment with SDG centroids therefore uses a benchmark that is itself politically negotiated and internally inconsistent — not a neutral standard. Gibbons et al. (1994) distinguish Mode 1 knowledge production, where problems are set and solved within academic disciplinary interests, from Mode 2, where problems arise in contexts of application; to the extent that research follows Mode 1, its agendas may reflect internal disciplinary dynamics rather than external policy needs. This observation also operates alongside a further ambiguity: what it means to be "aligned" with a framework whose component goals are in tension with each other, and it suggests that divergence is not uniformly problematic — where research explores possibilities that policy discourse has not yet absorbed, distance may reflect productive exploration rather than failure of alignment.

The two-communities theory (Caplan, 1979) holds that researchers and policy makers live in separate worlds with different values, reward systems, and languages, so divergence is the default state of research-policy interaction rather than a failure. Taken together, this literature indicates that the divergence between AI research and sustainability policy is a substantive feature of the two knowledge systems rather than an artefact of measurement. Existing studies document the gap through coverage, citation, and impact-based methods, while theory explains it through legitimacy, expert communities, and the structural divide between academic and policy communities. None of these studies measures semantic framing within a single SDG. This study measures how close research and policy texts lie in semantic space; that closeness is a first step, not proof of alignment.

2.4 Semantic Methods: Rationale and Cautions

This section explains the method behind this study and why it needs care.

The use of Sentence-BERT (Reimers & Gurevych, 2019) and related transformer-based encoders for semantic comparison has been validated across several domains relevant to this study, but applying them to compare two distinct discourse corpora — research abstracts and policy documents — is a novel usage rather than a directly benchmarked one. This method follows the distributional hypothesis that meaning can be described through patterns of linguistic co-occurrence, a principle whose modern computational implementations trace back to Harris (1954)'s foundational observation that the distribution of a linguistic element is "the sum of all its environments." Sentence-BERT and related transformer-based encoders are therefore used here as pragmatic tools for large-scale semantic comparison, while the interpretation of distance remains descriptive rather than causal. Gjorgjevikj et al. (2025) benchmarked sentence encoders specifically for SDG association tasks, finding that general-purpose models from the sentence-transformers family perform competitively as baselines, and that domain-specific fine-tuning further improves predictive performance while reducing sensitivity to description length.

Rodriguez et al. (2023)'s conText framework provides a closer computational-social-science precedent for comparing meaning across contexts. Their embedding-regression approach estimates how focal terms are used differently across covariates such as party, time, or corpus, while explicitly treating embedding-based "meaning" as a thin distributional description rather than a causal model of human interpretation. This study does not adopt their word-level regression estimator, but it follows the same broad premise that semantic representations can be used descriptively to compare how different discourse communities frame shared terms or categories.

McCarthy et al. (2026) supply a yet closer empirical precedent for the present design. Applying a multi-target SciBERT classifier to Ross Sea MPA research and policy-prioritised topics, they find that 59% of policy-prioritised research topics appear in fewer than 15 papers, showing that research–policy coverage gaps are measurable with transformer encoders outside the AI–SDG setting. This study extends that logic from a single conservation domain to the full SDG landscape, and from coverage to within-SDG semantic framing.

An older corpus-linguistic tradition gives a further reason for caution when interpreting cross-corpus semantic distance. Biber's multidimensional analysis of spoken and written English showed that register differences are not simple binary contrasts, but continuous patterns of co-occurring linguistic features across genres (Biber, 1988). For the present study, this matters because research abstracts and policy documents may differ not only in substantive topic, but also in abstraction, modality, institutional stance, and rhetorical function. Embedding distance is therefore treated as a measure of observed textual-semantic proximity, not as a clean measure of topic alone.

Embedding models let us compare research and policy texts, but applying them across two corpora is new. They represent meaning through distribution, not cause, and cannot separate topic from register. The measures used here are descriptive: they report how close or distant the two corpora are in semantic space, not whether that closeness is real alignment.

2.5 Research Gap

To the best of my knowledge, the literature reviewed above has not produced: (i) a systematic comparison of how academic AI-for-sustainability research and institutional SDG policy discourse differ in SDG coverage and within-SDG semantic framing — prior work measures research coverage, citation patterns, or policy-document uptake, but does not compare the two corpora directly on these dimensions; or (ii) an examination of whether coverage gaps and semantic gaps are positively related, weakly related, or inversely related. These are the contributions of the present study, and they answer the central research question set out in the Introduction. Together, they provide a systematic framework for distinguishing research-policy coverage from research-policy semantic proximity within a shared SDG representation. This framework is deliberately diagnostic: it maps where observed divergence is visible, without classifying whether any given gap reflects productive or problematic disconnection — the latter would require qualitative follow-up beyond the scope of embedding analysis.

3 Methodology

3.1 Research Corpus

This study uses only publicly available text data and code. It does not involve human participants or personal data. No ethical approval was required.

The research corpus consists of 3,105,144 English-language research abstracts retrieved from the OpenAlex API (Priem et al., 2022), covering publications between 2018 and 2025. Retrieval combined OpenAlex's native SDG work filter with four high-precision AI search terms (`artificial intelligence`, `machine learning`, `deep learning`, `neural network`), giving 68 queries (17 SDGs $\times$ 4 terms). The term set follows established practice for delimiting AI corpora (Hellwig et al., 2019; OECD, 2024); because the aim is macro-level semantic alignment rather than exhaustive publication counting, precision is prioritised at the retrieval stage to keep non-AI noise out of the embedding space. Complete query parameters are documented in the Supplementary Methodology (Appendix A).

A direct test of this scope condition re-ran retrieval using field-of-study membership for the same two AI fields (OECD, 2024) and re-scored the resulting 50,000-paper corpus through the identical pipeline. The concept-retrieved coverage and semantic-gap rankings correlate strongly with the keyword baseline (SDG-share Spearman $\rho = 0.937$, coverage-gap rank $\rho = 0.88$, semantic-gap rank $\rho = 0.88$; Appendix G), so the reported divergence patterns are not an artefact of the four-term query choice.

An important asymmetry follows: the research corpus is a tight intersection of AI and SDG content (AI $\cap$ SDG), while the policy corpus (Section 3.2) is a broader union covering institutional SDG discourse and AI governance. Between-SDG rankings are conditional on this asymmetric construction; following Armitage et al. (2020), the retrieval universe is a method-dependent corpus boundary, not a neutral population of all SDG-relevant AI research. After deduplication and removal of records lacking usable abstracts, 3,105,144 abstract-derived text segments were retained. No balancing by SDG or citation count is applied, because trimming would distort the coverage profile the study aims to measure.

3.2 Policy Corpus

The policy corpus contains 40,597 text segments from three collections, representing 6,367 unique source documents. It should be read as mixed *international institutional SDG policy discourse* rather than the full universe of AI sustainability policy:

1. **Curated AI and SDG policy documents** (13,123 segments, 94 documents): UN, OECD, EU, UK, and AU AI governance frameworks; IPCC assessment reports; SDSN sustainable development reports; and national AI strategies. These are AI-governance instruments by construction, forming the innovation-facing slice of the corpus.
2. **SDGi Voluntary National Review and Voluntary Local Review corpus** (SDGi, 2024) (21,346 segments, 4,225 documents): National and local self-assessments submitted to the UN High-Level Political Forum. The creators describe it as the most comprehensive multilingual collection of SDG-labelled texts to date, spanning all 17 goals and centred on SDG implementation rather than any single sector.
3. **UN General Debate Corpus** (Harvard Dataverse, 2024) (6,128 segments, 2,048 documents): SDG-relevant passages from General Debate speeches, sessions 70–80 (2015–2024). Its creators analyse SDG coverage directly and report climate action (SDG 13), peace and governance (SDG 16), and partnerships (SDG 17) among the most prominent themes, an orientation the analysis below also observes.

After cross-source deduplication by exact text match (SDGi retained first), 40,597 unique segments remained. Segment identifiers encode source document identity for document-level weighting. The three collections differ in genre and scope: the curated subset is AI-governance-focused, while SDGi and UNGDC reflect broader institutional SDG discourse. The full-corpus policy distribution should be read accordingly; the curated sub-corpus (94 documents) is the more relevant reference for AI-governance-specific interpretation. Appendix H.3 reports a source-family sensitivity check. Segmentation is token-aware and detailed in the Supplementary Methodology (Appendix A).

3.3 Embedding Model and Normalisation

All texts were encoded with `all-mpnet-base-v2` (Sentence-Transformers, 2021) from the sentence-transformers library (Reimers & Gurevych, 2019), producing 768-dimensional dense vectors. This pre-trained model requires no fine-tuning on the data under analysis, so the embedding space is a stable, pre-existing coordinate system. Section 4.6 tests sensitivity to this choice against a domain-specialised scientific encoder of the same dimensionality. All embeddings were L2-normalised to unit vectors so that cosine similarity equals their dot product. A truncation issue discovered during development was corrected by enforcing the token budget at segmentation time; the fix and its verification are described in the Supplementary Methodology (Appendix A).

3.4 Supervised Reference Classifier

The canonical instrument is a supervised multinomial logistic regression (LR) classifier trained on pooled single-label reference data from five independent annotated corpora:

- **OSDG Community Dataset** (Pukelis et al., 2022) (30,532 texts, SDGs 1–16): crowdsourced excerpts from reports, policy documents, and abstracts, each validated by multiple volunteers.
- **SDG Classification Benchmark** (SDG Classification Expert Group, 2025) (281 expert-verified texts, SDGs 1–17): short text snippets annotated by multiple domain experts with consensus resolution.
- **IISD SDG Knowledge Hub** (Wulff et al., 2023) (6,122 texts, SDGs 1–17): IISD journalism and policy commentary on 2030 Agenda implementation.
- **SDGi Corpus** (SDGi, 2024) (20,267 texts, SDGs 1–17): government Voluntary National and Local Reviews submitted to the UN.
- **Aurora dataset** (Vanderfeesten et al., 2020) (4,971 expert-validated texts, SDGs 1–17): academic abstracts validated by domain researchers.

All five corpora supply human- or expert-validated labels rather than machine-generated ones, and they differ in provenance: OSDG is a crowdsourced volunteer-confirmed collection, SDGi compiles government reviews, the Knowledge Hub is journalism, and Aurora is academic expert-validated text. The mix broadens the label base, but OSDG alone contributes about 49% of all labelled texts and draws on UN-adjacent documents, so the reference data remain weighted toward policy vocabulary. No high-quality graded relevance sets exist for SDG classification, making this combined, multi-source collection the best available resource (Kashnitsky et al., 2024).

The pooled reference data (n=62,173 after single-label filtering) were split into training and held-out test sets (n=52,835 train, n=9,338 test) by stratified document-grouped splitting: all texts from the same source document are kept in the same fold, enforcing strict separation between training and evaluation. The champion LR ($C = 10.0$, L2 penalty, `class_weight=None`, solver `lbfgs`) was selected by 5-fold GroupKFold cross-validation on the training pool and reaches macro-$F_1 = 0.817$ on the held-out test set (Section 4.1). The grid-search protocol, hyperparameter rationale, and the MLP robustness comparison are reported in Appendix E. The validation task does not evaluate a rejection option for non-SDG texts; this is acceptable because the research corpus is first restricted to an OpenAlex AI-and-SDG universe, so the classifier answers a relative question among the 17 SDGs.

3.5 Scoring and Assignment

A text is assigned to the SDG with the highest predicted probability from the trained LR classifier (argmax over 17 classes). Every assignment is traceable to 768 signed logistic-regression coefficients per SDG class, a transparent decision boundary with no hidden layers, no dropout, and no random-seed sensitivity. Macro-F_1, the unweighted mean of per-SDG scores, is the primary validation metric (Section 4.1); LR also provides calibrated per-class probabilities used as a soft-assignment comparison against the canonical hard argmax.

3.6 Classifier Calibration Across Discourse Types

Because the classifier is trained on pooled reference data that are themselves policy-adjacent (Section 3.4), the scoring instrument behaves differently on the two corpora. Policy segments receive systematically higher LR predicted probabilities than research abstracts (mean 0.769 vs 0.663, a 0.106 gap). The gap has two causes the data cannot separate: a genre artefact (the reference data skew toward policy vocabulary) or a substantive difference (policy texts address SDG implementation more directly). Either way the asymmetry does not threaten the findings: hard argmax assignment uses only within-corpus ranking, not absolute probability levels, so the directional gap does not reorder goals. The asymmetry is not an SDGi circularity effect: the training set and the scored policy corpus are disjoint by construction.

The combined PCA landscape (Figure 2) visualises this asymmetry: the policy cloud hugs the 17 reference-classifier centroids more tightly than the research cloud. The projection is descriptive only (the first two components explain 9.2% and 3.7% of variance); all reported cosine distances remain in the original 768-dimensional space.

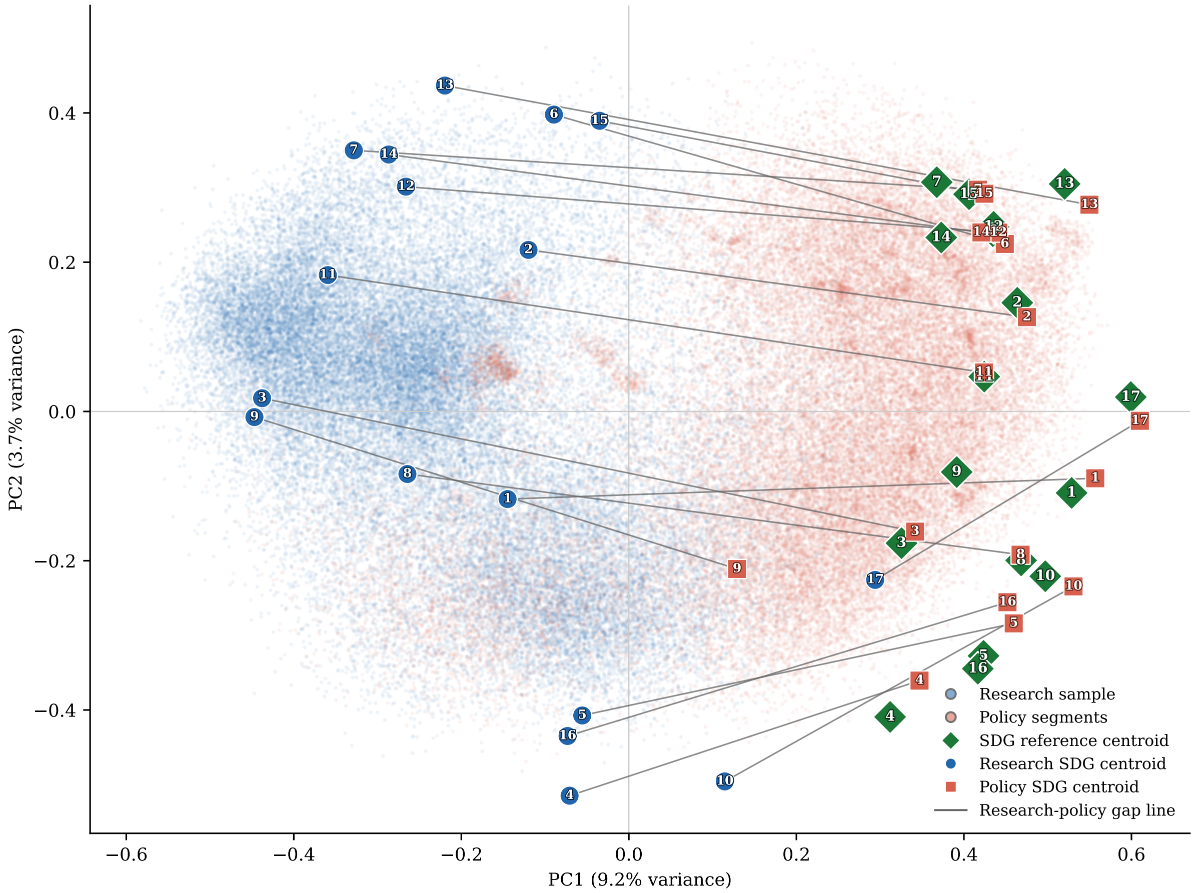

Figure 2. PCA semantic landscape of the combined research–policy embedding space. Descriptive two-dimensional projection only; formal semantic distances are computed in the original embedding space.

3.7 Coverage Gap Analysis

For each corpus item, the predicted SDG is the LR argmax (hard assignment, Section 3.5); the coverage profile is the proportion of items assigned to each SDG.

The policy corpus is not a flat set of documents: its 40,597 segments are grouped into 6,367 source documents of very different lengths (SDGi: 21,346 segments across 4,225 documents; curated AI/SDG: 13,123 across 94; UNGDC: 6,128 across 2,048). A segment-weighted profile would let a handful of long reports dominate the count, so the analysis instead computes a *document-weighted* profile in which every source document counts equally regardless of length. Concretely: (1) the segment score vectors of each document are averaged into one document-level vector; (2) that vector is hard-assigned to its top SDG;

(3) the coverage profile is the share of documents, not segments, assigned to each SDG. The research corpus uses the same logic at the paper level, where each of the 3,105,144 abstract texts is already a single unit. Document-weighted profiles are the canonical representation; segment-weighted profiles are retained as a diagnostic comparison.

The coverage gap for SDG j is defined as:

$$\mathrm{CoverageGap}_j = \left|\mathrm{ResearchCoverage}_j - \mathrm{PolicyCoverage}_j^{\mathrm{docwt}}\right|$$

The signed version of the same quantity is:

$$\mathrm{SignedCoverageGap}_j = \mathrm{ResearchCoverage}_j - \mathrm{PolicyCoverage}_j^{\mathrm{docwt}}$$

A positive value means research dominates SDG j; a negative value means policy dominates. These two definitions, together with research coverage and policy coverage, are the four predictors reported in Table 2.

3.8 Semantic Gap Analysis

The semantic gap measures within-SDG divergence: given that both corpora have texts assigned to SDG j, how differently do they discuss it? For each SDG j, the analysis collects all assigned research papers and all assigned policy segments (capped at 50 per source document to prevent single-document dominance; SDSN reports alone contribute up to 3,179 segments otherwise), computes the L2-normalised mean embedding of each cluster ($\mathbf{c}_j^{\mathrm{res}}$, $\mathbf{c}_j^{\mathrm{pol}}$), and defines:

$$\mathrm{SemanticGap}_j = 1 - (\mathbf{c}_j^{\mathrm{res}} \cdot \mathbf{c}_j^{\mathrm{pol}})$$

A gap of 0 indicates perfectly overlapping semantic directions; a gap approaching 1 indicates orthogonal framing.

The *per-document segment cap* of $K = 50$ (approximately 7,500 words) uses seeded random sampling, hence reproducible, and sits well above the median of roughly 14 segments per document. Repeating the computation at a cap of 20 and with no cap leaves the SDG gap rankings near-identical (rank correlation $\rho = 0.969$; Table 3, Section 4.5), so the findings are robust whether or not the cap is applied.

The semantic gap is not a clean measure of substantive disagreement. Following Biber and Conrad's distinction between register, genre, and style (Biber & Conrad, 2009), I treat the research–policy contrast primarily as a register contrast: the two corpora differ in situation of use, audience, and communicative purpose, not merely in topic. Embedding distance therefore mixes genuine within-SDG substantive divergence with register divergence in abstraction, modality, and institutional stance (Biber, 1988). The raw centroid distance is retained as the primary measure because it directly corresponds to the observed framing difference. Appendix F reports a register-adjusted sensitivity analysis: subtracting one global research–policy direction compresses the gaps modestly, while the stratified iterative variant reorders individual SDGs, most strikingly SDG 17, which flips from the smallest raw gap to the largest adjusted gap. These adjusted estimates bound the register-sensitive component of the gap without replacing the raw estimate.

3.9 Coverage–Semantic Interaction

The first hypothesis (H1) decomposes into four sub-hypotheses, each correlating a distinct coverage predictor with the within-SDG semantic gap across the 17 SDGs: H1a, coverage gap $\leftrightarrow$ semantic gap; H1b, research–policy dominance $\leftrightarrow$ semantic gap; H1c, research coverage $\leftrightarrow$ semantic gap; and H1d, policy coverage $\leftrightarrow$ semantic gap. H1a is the primary, motivating hypothesis. I compute Pearson r and Spearman ρ between each predictor and semantic-gap scores, with sensitivity analyses excluding SDG 4 (Section 3.10). Given $n = 17$, the tests have limited power to detect anything but a large linear effect; this is a scope condition.

3.10 Key Assumptions and Mitigations

Two design-level choices shape the comparison before any measurement takes place; both are methodological decisions rather than empirical findings.

Text-unit handling is a design choice, not a result. Research papers are single abstract-length texts (mean ∼200 words); policy items are 150–300-word segments from documents of highly variable length. Document-weighting and per-document segment caps are adopted up front to keep the two corpora comparable; they are fixed procedures, not findings derived from the data.

Labelled-data limitation on SDG 4. The research coverage assigns 14.6% of papers to SDG 4 (Quality Education, see later Section 4.2), flagged as artefact-affected because ML papers share core vocabulary ("learning", "training", "model") with the OSDG education corpus, a case of lexical brittleness in keyword-based semantic mapping (Armitage et al., 2020). Appendix B.1 confirms that SDG 4-assigned texts contain education-specific terms more often than matched non-SDG4 samples, but co-occur with ML-learning vocabulary. The result is retained as a learning-related SDG 4 assignment rather than a clean education estimate. Sensitivity analyses excluding SDG 4 are provided for all correlation results.

3.11 Methodology Summary

Figure 3 summarises the end-to-end methodology. All three tracks share preprocessing (cleaning, English filtering, 20-word minimum) and token-aware segmentation (margin 10, min 20 words). The reference track pools 62,173 single-label texts (52,835 train, 9,338 test) to train the LR classifier ($C = 10$, L2, macro-$F_1 = 0.817$); the research and policy tracks segment 3,105,144 abstracts and 40,597 policy segments respectively. All tracks are embedded with `all-mpnet-base-v2` (768-D, L2-normalised) and scored via argmax over 17 SDG classes.

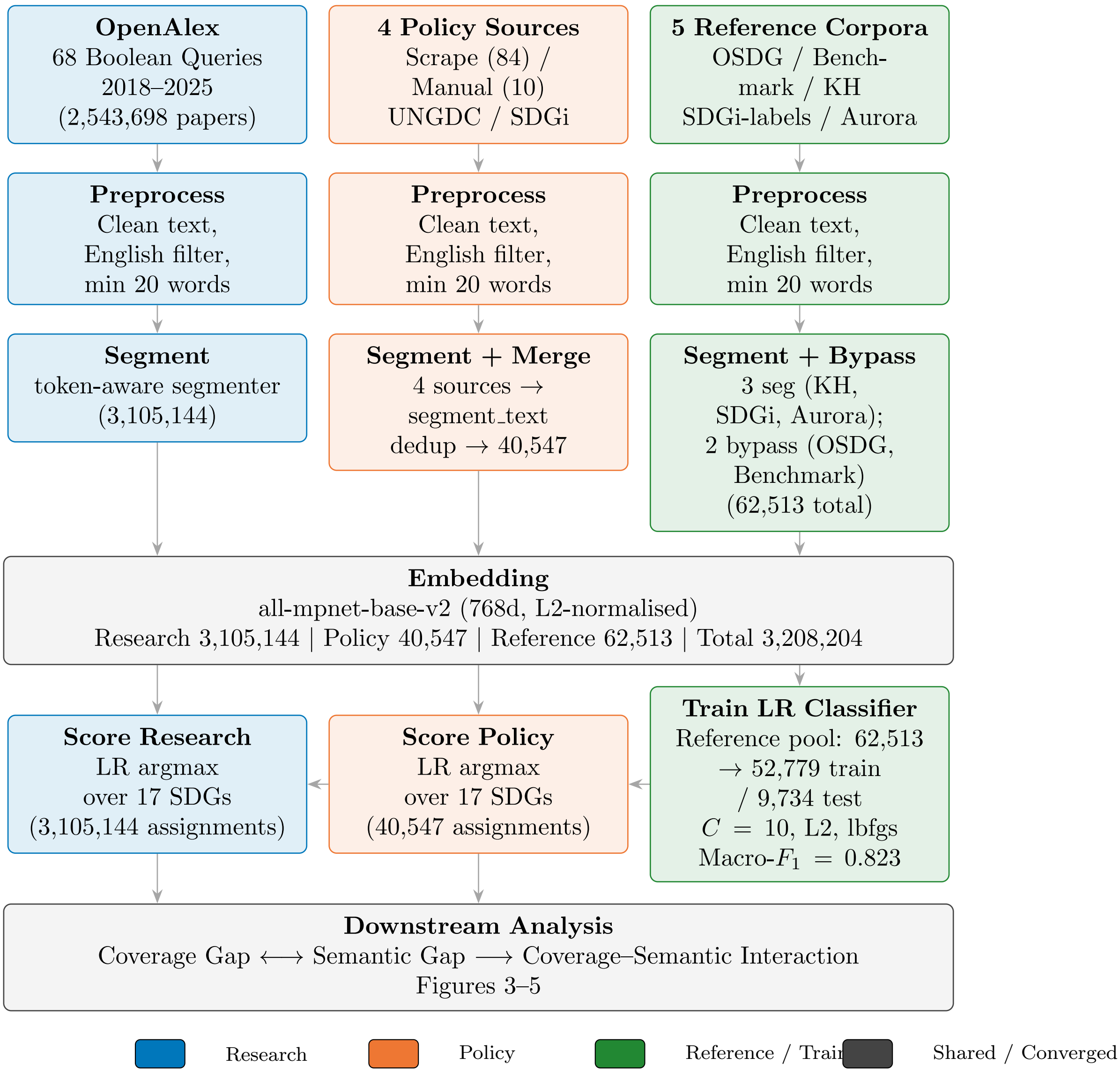

Figure 3. End-to-end methodology pipeline in three parallel data tracks. All tracks pass through preprocessing and token-aware segmentation; OSDG and Benchmark in Track 1 bypass segmentation (short single-label texts). Policy segments four sources then merges (40,597). Research segments all abstracts (3,105,144). The trained LR classifier scores both Research and Policy embeddings for downstream gap analysis.

4 Results

This section reports what the instrument measured: the validation scores, the coverage and semantic gaps, the H1 correlation tests, and the robustness of those rankings. It describes the findings; it does not interpret them. Interpretation, limits, and implications follow in the Discussion (Section 5).

4.1 Supervised Reference Classifier Validation

The LR classifier was evaluated on the held-out test set (n=9,338, document-grouped to prevent source leakage). It reaches macro-F1 = 0.817 and micro-F1 = 0.8309, far above the 1/17 random baseline (a 17-class random guess scores $1/17 \approx 0.059$). Per-class F1 values from the test set are reported in Table 1.

Table 1. LR test-set F1 scores (n=9,338).

SDG	LR test F1
SDG 1 (No Poverty)	0.747
SDG 2 (Zero Hunger)	0.825
SDG 3 (Good Health)	0.903
SDG 4 (Quality Education)	0.887
SDG 5 (Gender Equality)	0.877
SDG 6 (Clean Water)	0.871
SDG 7 (Clean Energy)	0.866
SDG 8 (Decent Work)	0.708
SDG 9 (Industry & Infra.)	0.770
SDG 10 (Reduced Inequalities)	0.613
SDG 11 (Sustainable Cities)	0.773
SDG 12 (Responsible Cons.)	0.780
SDG 13 (Climate Action)	0.833
SDG 14 (Life Below Water)	0.895
SDG 15 (Life on Land)	0.861
SDG 16 (Peace & Justice)	0.882
SDG 17 (Partnerships)	0.796
Macro-F1 (SDGs 1–17)	**0.817**

Notes: LR test F1 on the held-out pooled test set (all reference sources). The table shows the canonical 17-way classifier used throughout the main analysis.

The LR achieves consistently strong per-SDG performance: SDG 3 (F1 = 0.903), SDG 14 (0.895), and SDG 4 (0.887) are the most confidently classified (each above 0.89). The weakest classes are SDG 10 (Reduced Inequalities, F1 = 0.613) and SDG 8 (Decent Work, F1 = 0.708), consistent with their known semantic breadth and the overlapping cluster structure of adjacent goals (see centroid-similarity heatmap, Figure 7). Most goals score between 0.76 and 0.89, indicating reliable discrimination across the 17-class space. SDG 17 (F1 = 0.796) is no longer the weakest link — the supervised decision boundary separates partnership discourse more cleanly than the centroid average did — confirming that the goal's broad mandate resists compact semantic definition regardless of method.

The MLP robustness check (Appendix E) achieves a comparable test macro-F1 (MLP: 0.830; LR: 0.817), confirming that architecture choice is not the binding constraint on classification performance. The gap between LR and MLP is modest and does not justify the complexity trade-off — LR provides 768 signed coefficients per class, each directly attributable to a semantic dimension, while MLP's hidden layers diffuse this information across non-linear transformations.

In summary, the LR classifier separates the 17 SDGs with consistent and interpretable performance. The weakest classes — SDG 10 and SDG 8 — are known to sit in dense centroid clusters, and SDG 17, while no longer the lowest-F1 goal in the LR test set, remains the most method-sensitive SDG in the cross-sensitivity analysis. These scores mark a ceiling for the embedding-plus-linear-classifier approach; MLP results confirm that the ceiling is not an artefact of linear separability assumptions.

4.2 Coverage Gap: What SDGs Each Corpus Prioritises

Figure 4 presents the document-weighted policy coverage profile alongside the research coverage profile. The two profiles are markedly different.

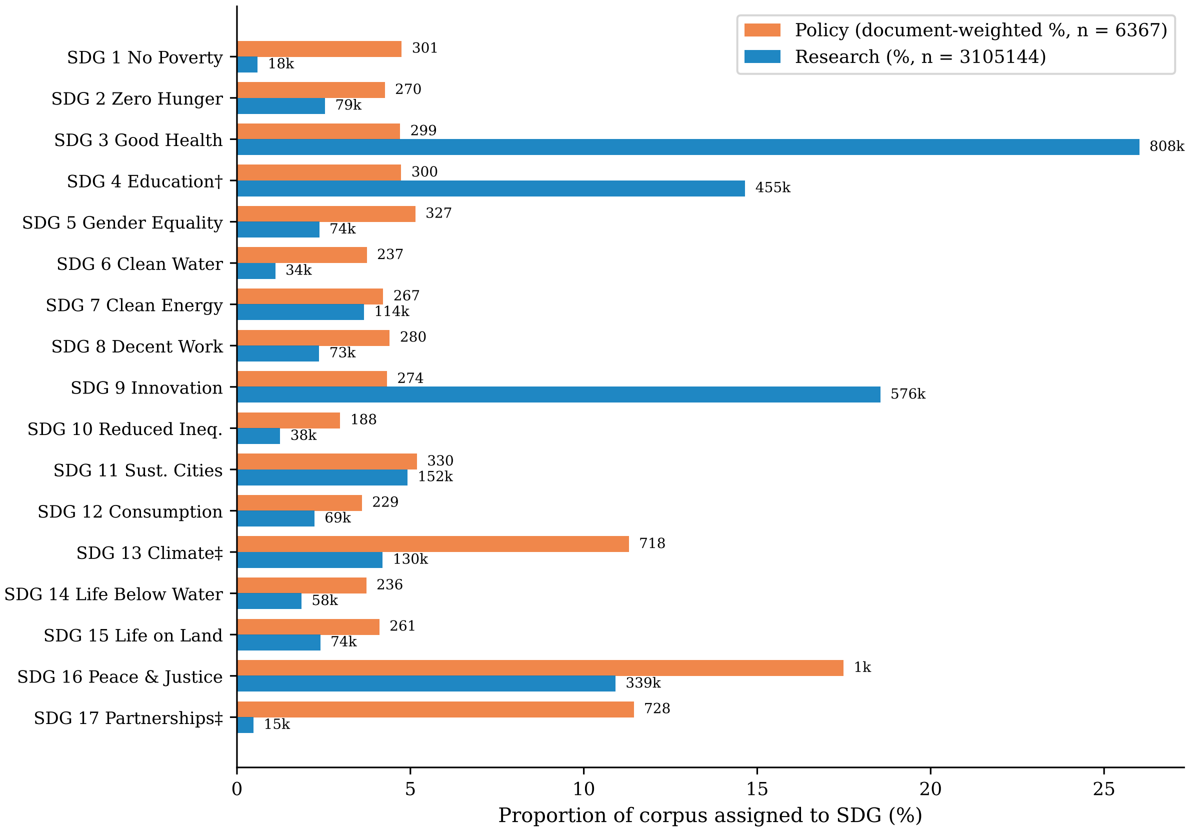

Figure 4. Coverage profiles by SDG.

Policy corpus. Three SDGs dominate the document-weighted profile of the full institutional policy corpus: SDG 16 (Peace, Justice and Strong Institutions, 17.5%), SDG 13

(Climate Action, 11.3%), and SDG 17 (Partnerships for the Goals, 11.4%), accounting for 40.2% of policy documents. This ordering is consistent with the literature review's account of SDG 17 as the cross-cutting means-of-implementation goal whose discourse concerns coordination and resourcing rather than a substantive topic (Griggs et al., 2017; Le Blanc, 2015); the climate–governance emphasis also matches the SDGi and UNGDC policy components described in Section 3.2. The SDG 16 lead is notable: under the supervised classifier, governance and justice discourse emerges as the most prevalent policy theme, ahead of both climate and partnerships. Appendix H.3 shows that this full-corpus profile is source-family sensitive: the curated AI/SDG sub-corpus is much more innovation-heavy, while the broader SDGi and UNGDC components are more climate-, partnerships-, and governance-oriented.

Research corpus. The research profile is broader, but it still concentrates strongly on a smaller set of SDGs. The largest research shares are SDG 3 (Good Health, 26.0%), SDG 9 (Innovation, 18.5%), and SDG 4 (Quality Education, 14.6%), followed by SDG 16 and SDG 11 at 10.9% and 4.9% respectively. The SDG 3 lead corroborates the literature review, where SDG 3 is the one goal every major AI and sustainability bibliometric study finds dominant — Bautista-Puig et al. (2021), Singh et al. (2024), Cowls et al. (2021), and Nedungadi et al. (2024) all rank it first, with SDG 9 and SDG 7 typically next. The SDG 4 share is the one point of disagreement: prior inventories do not rank SDG 4 among the leaders, so its third-place finish here is the same result flagged as likely inflated by machine-learning vocabulary (Section 3.10); read cautiously, the research profile agrees with the literature on SDG 3 and SDG 9 but over-states SDG 4. Even so, the research corpus clearly does not mirror the policy-side dominance of SDGs 16, 13, and 17.

Largest coverage mismatches. The five largest absolute coverage gaps are SDG 16 (0.066), SDG 3 (0.213), SDG 17 (0.110), SDG 13 (0.071), and SDG 9 (0.142). SDG 16 is now the most asymmetric goal — policy discourse strongly emphasises governance and institutions while research barely touches it — followed by the familiar research-over-policy dominance in health (SDG 3).

Taken together, three observations support reading the coverage profiles as a real signal rather than a measurement artefact. First, the instrument reaches macro-F1 = 0.817 on the held-out benchmark (Section 4.1), far above the 1/17 random baseline, so assignments are not random. Second, the research-side ranking reproduced here — SDG 3 dominant, with SDG 9 and SDG 7 next — matches the ordering reported independently by four prior bibliometric studies of AI and sustainability research Bautista-Puig et al. (2021), Singh et al. (2024), Cowls et al. (2021), Nedungadi et al. (2024) (Section 2), which used keyword and citation methods rather than embeddings. Third, this is all while the classifier is trained on reference corpora that are themselves policy-adjacent (Section 3.4). To summarize, that the research-covered SDG ranking nonetheless converges with independent, methodologically unrelated literature is the strongest available evidence that the classifier anchors a meaningful semantic coordinate system, not a policy-specific artefact.

4.3 Semantic Gap: How Differently Research and Policy Discuss Each SDG

Figure 5 shows the within-SDG semantic gap for all 17 SDGs. The gap ranges from 0.216 (SDG 17) to 0.481 (SDG 13), a span of 0.265 cosine units. A large gap means research and policy use different words and concepts for the same goal; a small gap means they frame it the same way. The largest gap is about 4.5 times the policy–research score gap (0.106, Section 3.6). SDG 13 has the largest gap, followed closely by SDG 3 and SDG 11. The ranking is stable when the segment cap is 20 per document or removed (rank correlation $\rho = 0.969$; Table 3), so the result does not depend on the sampling parameter.

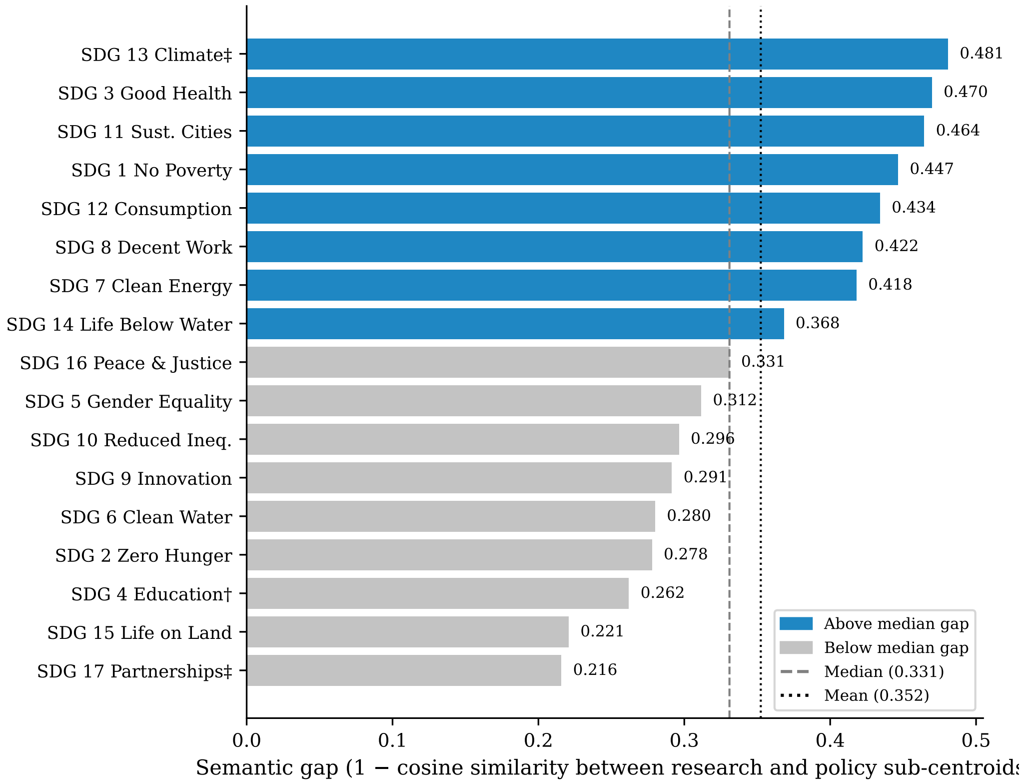

Figure 5. Within-SDG semantic gap by SDG, sorted descending. Dashed line = median (0.331); dotted line = mean (0.352). Dark bars exceed the median; light bars fall below it.

Largest semantic gaps. SDG 13 (Climate Action, 0.481), SDG 3 (Good Health, 0.470), SDG 11 (Sustainable Cities, 0.464), SDG 1 (No Poverty, 0.447), and SDG 7 (Clean Energy, 0.418) show the widest gaps. The top-gap SDGs are a mix of climate, health, and infrastructure goals where research and policy discourse diverge most sharply. This ordering is stable across the segment-cap sensitivity check (Table 3). Notably, SDG 17 (Partnerships, 0.216) now shows the smallest semantic gap — the supervised decision boundary separates partnership discourse more cleanly than the old centroid method, moving it from the most to the least divergent goal.

Smallest semantic gaps. SDG 17 (0.216), SDG 15 (Life on Land, 0.221), SDG 4 (Quality Education, 0.262), SDG 6 (Clean Water, 0.280), and SDG 9 (Industry and Infrastructure, 0.291) show the narrowest gaps, meaning research and policy frame these goals in close terms. A narrow gap does not mean balanced attention: SDG 9 is still research-dominant, SDG 17 is policy-dominant, and SDG 4 carries the learning-vocabulary concern (Appendix B.1). Appendix C.1 shows example texts for high- and low-gap cases.

4.4 Coverage–Semantic Interaction: Two Distinct Dimensions of Observed Divergence

Table 2 reports the H1a–H1d correlation tests between coverage measures and semantic gap scores across the SDGs.

The four predictors are research coverage, policy coverage, the absolute research–policy coverage gap, and the signed research–policy dominance. Each panel shows the result across all 17 SDGs together with tests dropping SDG 4 and SDG 17.

Table 2. H1a–H1d correlation tests between coverage predictors and the within-SDG semantic gap.

Config	Pearson r	p	Spearman ρ	p
H1a. Coverage gap ↔ Semantic gap				
Primary observed SDGs (n=17)	-0.000	0.999	-0.078	0.765
Excluding SDG 4 (n=16)	0.055	0.840	0.018	0.948
Excluding SDG 17 (n=16)	0.109	0.689	0.047	0.863
H1b. Policy–research dominance ↔ Semantic gap				
Primary observed SDGs (n=17)	0.128	0.626	0.054	0.837
Excluding SDG 4 (n=16)	0.229	0.393	0.153	0.572
Excluding SDG 17 (n=16)	-0.011	0.966	-0.135	0.617
H1c. Research coverage ↔ Semantic gap				
Primary observed SDGs (n=17)	0.123	0.637	0.216	0.406
Excluding SDG 4 (n=16)	0.220	0.412	0.341	0.196
Excluding SDG 17 (n=16)	0.055	0.841	0.059	0.829
H1d. Policy coverage ↔ Semantic gap				
Primary observed SDGs (n=17)	-0.030	0.909	0.159	0.541
Excluding SDG 4 (n=16)	-0.052	0.849	0.215	0.425
Excluding SDG 17 (n=16)	0.134	0.621	0.368	0.161

The evidence does not support H1a: across the 17 SDGs, coverage divergence and semantic divergence show no detectable association ($r = -0.000$, $p = 0.999$; $n = 17$). Given the small number of SDG-level observations, this should be interpreted as evidence against a strong monotonic relationship rather than evidence of exact independence. H1c likewise shows no detectable association: research coverage and semantic gap are uncorrelated ($r = 0.123$, $p = 0.637$). Two exploratory tests — H1d, policy coverage ($r = -0.030$, $p = 0.909$), and H1b, signed dominance ($r = 0.128$, $p = 0.626$) — are also near-zero. All correlations remain small and non-significant under both exclusions; however, H1a and H1b flip sign when SDG 17 is dropped (Table 2), consistent with SDG 17's broader sensitivity to measurement configuration (§4.5).

Figure 6 shows each SDG. The most striking pattern is the absence of a pattern: SDGs with large coverage gaps span the full range of semantic gaps (e.g. SDG 16 has the largest coverage gap but a below-median semantic gap, while SDG 3 has a large coverage gap and the second-largest semantic gap). Research attention coexists with both low- and high-gap cases (e.g. SDG 9 high-coverage low-gap, SDG 3 high-coverage high-gap), which is why H1c is near zero. No SDG drives a positive association; the null result is consistent across all configurations.

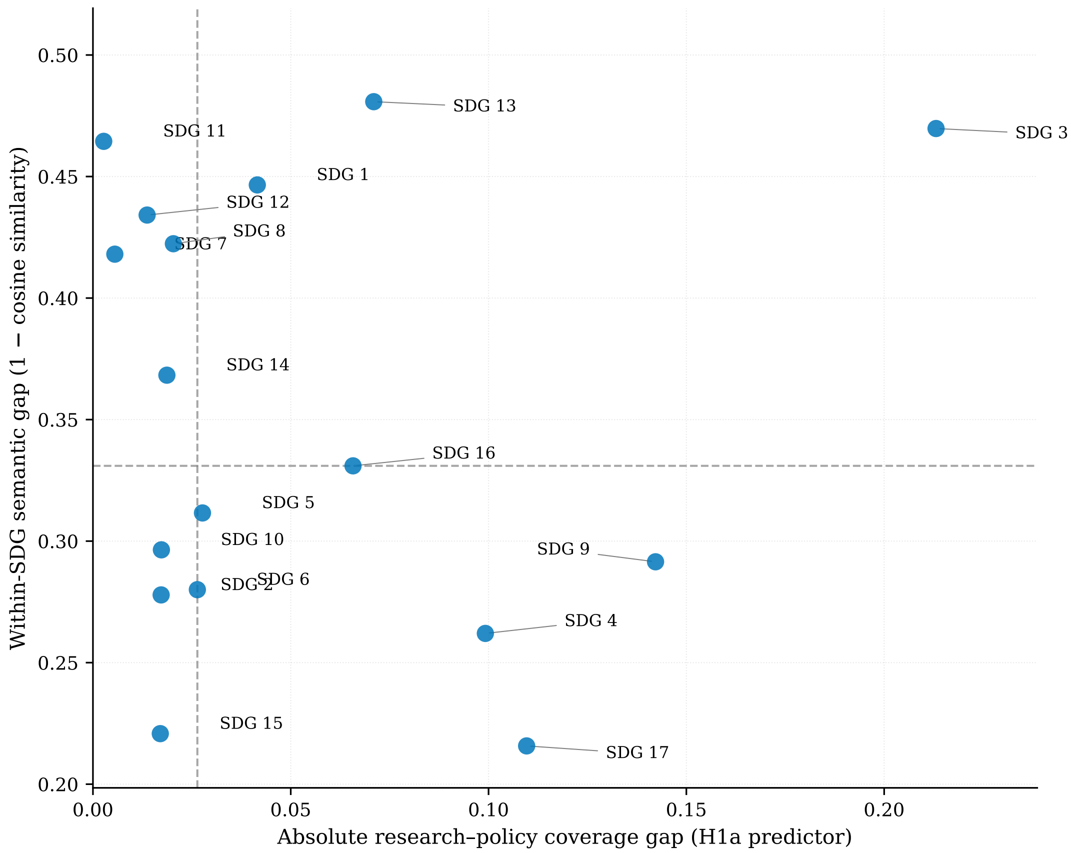

Figure 6. Coverage gap vs semantic gap, one point per SDG.

4.5 Cross-Sensitivity Robustness of the Gap Rankings

The semantic gap rankings discussed below are not equally stable across all measurement choices. Table 3 shows each SDG's gap rank (1 = largest gap) under the canonical MPNet—LR configuration and six alternatives spanning three axes: policy source family (curated AI/SDG, SDGi, UNGDC), segment cap (20, none), and retrieval strategy (concept-based field-of-study). The encoder and assignment-method axes are examined separately in Section 4.6 (Table 5).

Table 3. Cross-sensitivity robustness of within-SDG semantic-gap rankings across policy source, segment cap, and retrieval strategy.

SDG		Policy source			Segment cap		Retrieval	
	Canon	Curated	SDGi	UNGDC	20	None	LR	MLP
SDG 1	4	3	4	3	4	3	1	2
SDG 2	14	11	13	11	14	13	12	10
SDG 3	2	9	2	1	1	4	3	5
SDG 4	15	15	14	15	15	15	14	16
SDG 5	10	12	11	10	11	9	9	14
SDG 6	13	10	15	12	13	14	16	12
SDG 7	7	1	8	4	7	6	8	4
SDG 8	6	6	6	7	6	7	4	3
SDG 9	12	14	1	13	10	12	13	11
SDG 10	11	7	12	14	12	11	10	15
SDG 11	3	4	3	5	3	2	2	1
SDG 12	5	2	7	2	5	5	11	13
SDG 13	1	5	5	6	2	1	5	6
SDG 14	8	8	9	8	8	8	6	7
SDG 15	16	13	16	16	16	16	17	17
SDG 16	9	16	10	9	9	10	7	8
SDG 17	17	17	17	17	17	17	15	9
Rank Corr (ρ)	1.00	0.74	0.81	0.91	0.99	0.99	0.88	0.70

Notes: Each cell reports the within-SDG semantic-gap rank (1 = largest gap, 17 = smallest gap). The Canon column is the canonical MPNet—LR ranking. Policy-source columns compare the keyword-retrieved research profile against each policy-source family. Segment-cap columns compare cap 20 and no cap. The Retrieval column replaces keyword retrieval with concept-based (AI/ML field-of-study) retrieval. Rank Corr (ρ) is the Spearman correlation of each column's SDG gap ranks against the canonical column.

The same cross-sensitivity structure holds for the coverage gap (Table 4). The within-SDG coverage-gap ranking is stable across retrieval strategy ($\rho = 0.88$ for concept-based field-of-study retrieval versus the keyword baseline); coverage was also stable across embedding architectures (Section 4.6, Table 6).

4.6 Domain-Encoder Sensitivity (Same-Dimension Scientific Encoder)

The cross-sensitivity check above compares the canonical `all-mpnet-base-v2` (768-d) encoder against the lighter `all-MiniLM-L6-v2` (384-d) encoder. Because the two differ in both architecture and dimensionality, that comparison bounds encoder sensitivity only modulo the dimensionality drop. To isolate architecture and domain specialisation from dimensionality, this section adds a third encoder, `allenai/scibert_scivocab_uncased` (Beltagy et al., 2019), a scientific-domain BERT that is also 768-d. SciBERT is the domain-specialised counterpart to the general-purpose MPNet baseline (§3.3) and the architecture used by McCarthy et al. (2026) in the closest published precedent for the present design. MPNet and SciBERT therefore differ in architecture and training domain but share dimensionality, so any remaining disagreement between them is a pure architecture/domain effect. Both encoders embed the identical canonical MPNet-segmented texts, and SciBERT is scored on the same 50,000-paper representative research sample used by the sample-stability ladder, so the comparison holds inputs fixed.

The more revealing result is the gap magnitude. SciBERT gives a mean semantic gap of 0.05 (cosine 0.95) against MPNet's 0.35 (0.65): the research–policy distance collapses to near uniformity across all 17 SDGs. The embeddings are not degenerate — SciBERT's research and policy cohesion (0.88, 0.92) exceed MPNet's (0.40, 0.67) — but its scientific pretraining fuses the two genres. SciBERT's weights are tuned to 1.14M scientific papers, so policy text, which mirrors scientific vocabulary and formal syntax, lands in the same region as research abstracts; MPNet, trained on general web text, keeps the technical-method versus institutional-mandate contrast that SciBERT erases. A general-purpose encoder is therefore the right ruler: a domain-specialised one that fuses the genres cannot measure the register gap this study tracks.

Tables 5 and 6 report the within-SDG gap rankings under MPNet, MiniLM, and SciBERT for each assignment method. Despite the compressed magnitudes, the semantic-gap ranking is moderately preserved across the three encoders (MPNet–LR vs. SciBERT–LR Spearman $\rho = 0.39$; MPNet–LR vs. MiniLM–LR $\rho \approx 0.65$), and the coverage-gap ranking is highly stable (MPNet–LR vs. SciBERT–LR $\rho = 0.95$). The same SDGs lead and trail under every encoder: SDG 13 is the most divergent under every encoder, and SDG 17 the least under the supervised classifiers. The research–policy divergence findings are therefore not an artefact of the MPNet geometry or its 768-d dimensionality. The magnitudes are encoder-dependent — a caveat the study's choice of a general-purpose encoder was designed to meet — but the relative ranking and the invariance of coverage structure hold across general-purpose and domain-specialised encoders alike.

Assignment-method sensitivity within MPNet tells a similar story. Under the canonical encoder, the three assignment methods—LR, zero-shot nearest-centroid, and the MLP robustness check—show varied agreement (MPNet LR vs. MLP $\rho = 0.83$, LR vs. zero-shot $\rho = 0.45$, Table 5). SDG 7 moves from rank 7 under LR to rank 2 under MLP; SDG 5 moves from rank 10 to rank 16 between LR and zero-shot; SDG 8 moves from rank 6 to rank 15. SDG 17 is the most method-sensitive goal, sitting at rank 17 under LR but rank 7 under zero-shot. The LR baseline is therefore the canonical reference, with the MLP and zero-shot columns providing uncertainty bounds.

4.7 Robustness of the Semantic-Gap Ranking to the Distance Functional

The centroid gap (§4.3) is the cosine distance between per-SDG mean embeddings. A reviewer may object that a single mean discards the shape of the two point clouds: two SDGs could share a mean direction yet differ in spread or modality (Villani, 2008). This section tests that objection directly by recomputing the per-SDG research–policy gap under a battery of distribution-aware metrics on the identical point clouds, holding the supervised SDG assignment and embeddings fixed; only the distance functional changes.

The canonical gap is reproduced exactly as a validation gate: the full-corpus centroid gap equals the published value to 10^{-4}, and the streamed research means reproduce the committed centroids exactly. Full-corpus metrics (sliced Wasserstein (Bonneel et al., 2015), Chamfer (Borgefors, 1988), Gaussian-2-Wasserstein (Dowson & Landau, 1982), Grassmann (Edelman et al., 1998), linear MMD (Gretton et al., 2012)) are computed over all assigned papers and carry no sampling variance; sampled metrics (exact EMD (Rubner et al., 2000), Sinkhorn (Cuturi, 2013), energy (Székely & Rizzo, 2013), RBF-MMD (Gretton et al., 2012), C2ST (Lopez-Paz & Oquab, 2017)) use a 50,000-paper draw per SDG, with SDGs 1 and 17 using their full population.

Shape-sensitive metrics preserve the ranking. Gaussian-2-Wasserstein ($\rho = 0.843$), RBF maximum mean discrepancy ($\rho = 0.664$), exact optimal-transport EMD ($\rho = 0.907$), sliced Wasserstein ($\rho = 0.765$), and energy distance ($\rho = 0.772$; (Sejdinovic et al., 2013)) all preserve the canonical ordering at $p < 0.001$ (Spearman, 1904). SDG 13 is the most divergent SDG under every metric; its primacy is unconditional.

What centroids discard is quantified, not hidden. The Gaussian-2-Wasserstein distance decomposes into a mean term and a shape (covariance) term (Dowson & Landau, 1982; Heusel et al., 2017). Across SDGs the shape term accounts for a mean share of $\approx 57\%$ of the distance. Centroids therefore drop real distributional signal—but that signal does not reorder the SDGs, which is why the centroid ranking is robust.

Margin-level nuance. The second and third canonical positions (SDGs 3 and 11) are less stable: several distribution-aware metrics elevate SDGs 1, 7, 12, and 14 instead. This is a genuine margin-level reshuffle among SDGs with similar gap magnitudes, not an error; the overall monotonic agreement (positive ρ for every shape-sensitive metric) holds.

Separation, not ranking, from C2ST. The classifier two-sample test (Lopez-Paz & Oquab, 2017) returns AUC ≈ 1 for all 17 SDGs, so it confirms that research and policy are separable in every SDG but carries no ranking signal ($\rho \approx 0$). Its near-constant value is expected, not a failure.

Sampling stability. The sampled metrics use a 50,000-paper draw; a replicate seed changes any sampled gap by at most 0.007, confirming the estimate is stable and the default seed is representative. Together with the cross-sensitivity checks (§4.5), the centroid gap ranking is therefore robust to both the assignment method and the distance functional.

Table 4. Cross-sensitivity robustness of within-SDG coverage-gap rankings across policy source and retrieval strategy.

SDG		Policy source			Retrieval	
	Canon	Curated	SDGi	UNGDC	LR	MLP
SDG 1	7	17	4	11	6	8
SDG 2	13	8	10	12	9	11
SDG 3	1	3	1	2	1	2
SDG 4	4	4	3	6	11	17
SDG 5	8	11	8	17	7	7
SDG 6	9	15	7	15	8	12
SDG 7	16	6	16	8	17	15
SDG 8	10	12	9	9	14	13
SDG 9	2	1	2	5	2	1
SDG 10	12	14	13	14	12	9
SDG 11	17	5	14	7	15	5
SDG 12	15	13	12	10	16	16
SDG 13	5	7	17	4	5	6
SDG 14	11	16	15	16	10	10
SDG 15	14	10	11	13	13	14
SDG 16	6	9	6	1	4	3
SDG 17	3	2	5	3	3	4
Rank Corr (ρ)	1.00	0.40	0.74	0.52	0.88	0.56

Notes: Each cell reports the within-SDG coverage-gap rank (1 = largest gap, 17 = smallest gap). The Canon column is the canonical MPNet—LR ranking. Coverage gap = |research proportion − policy proportion| per SDG, using document-weighted policy proportions (Assumption A19). Policy-source columns compare the keyword-retrieved research profile against each policy-source family. The Retrieval column replaces keyword retrieval with concept-based (AI/ML field-of-study) retrieval. The Segment-cap axis is omitted: coverage gap is segment-cap-independent. Rank Corr (ρ) is the Spearman correlation of each column's SDG coverage-gap ranks against the canonical column.

Table 5. Domain-encoder sensitivity of within-SDG semantic-gap rankings across embedding architectures of differing domain specialisation but matched dimensionality.

SDG	Encoder (embedding architecture)						
	MPNet (768-d)			MiniLM (384-d)		SciBERT (768-d)	
	LR	MLP	ZS	LR	MLP	LR	MLP
SDG 1	4	3	8	7	4	7	3
SDG 2	14	12	10	12	14	13	15
SDG 3	2	5	4	4	2	1	7
SDG 4	**15**	**14**	**14**	**16**	**17**	**9**	**11**
SDG 5	*10*	*16*	*16*	*2*	*12*	*4*	*4*
SDG 6	13	9	6	15	13	11	1
SDG 7	**7**	**2**	**1**	**6**	**8**	**5**	**8**
SDG 8	6	7	15	3	1	17	16
SDG 9	**12**	**11**	**17**	**11**	**9**	**12**	**13**
SDG 10	*11*	*15*	*13*	*5*	*3*	*2*	*10*
SDG 11	*3*	*1*	*2*	*8*	*6*	*6*	*6*
SDG 12	*5*	*6*	*9*	*10*	*7*	*15*	*9*
SDG 13	**1**	**4**	**5**	**1**	**5**	**3**	**5**
SDG 14	*8*	*8*	*3*	*14*	*15*	*10*	*12*
SDG 15	*16*	*17*	*12*	*9*	*16*	*8*	*2*
SDG 16	*9*	*10*	*11*	*13*	*11*	*14*	*14*
SDG 17	**17**	**13**	**7**	**17**	**10**	**16**	**17**
Rank Corr (ρ)	1.00	0.83	0.45	0.65	0.71	0.39	0.26

Notes: Each cell reports the within-SDG semantic-gap rank (1 = largest gap, 17 = smallest gap) under that encoder and assignment method. MPNet (768-d) is the canonical general-purpose encoder; MiniLM (384-d) is a smaller general-purpose encoder; SciBERT (768-d) is a scientific-domain encoder. MPNet and SciBERT share dimensionality (768-d), isolating architecture/domain from the dimensionality drop that confounds the MPNet–MiniLM pair (Section 4.6). LR = canonical supervised logistic regression; policy segments capped at 50/doc/SDG. MLP = 4-layer/384-hidden network. Zero-shot = nearest-centroid on SDG reference centroids, reported for the canonical MPNet encoder only (scoped to a single supervised-vs-nearest-centroid comparison, Appendix H.2). Rank Corr (ρ) is the Spearman correlation of each column's SDG gap ranks against the canonical MPNet-LR column.

Table 6. Domain-encoder sensitivity of within-SDG coverage-gap rankings across embedding architectures of differing domain specialisation but matched dimensionality.

SDG	Encoder (embedding architecture)						
	MPNet (768-d)			MiniLM (384-d)		SciBERT (768-d)	
	LR	MLP	ZS	LR	MLP	LR	MLP
SDG 1	**7**	**7**	**6**	**7**	**7**	**6**	**6**
SDG 2	13	14	14	10	10	12	9
SDG 3	**1**	**1**	**2**	**1**	**1**	**1**	**1**
SDG 4	**4**	**4**	**3**	**5**	**3**	**3**	**4**
SDG 5	*8*	*8*	*8*	*12*	*8*	*9*	*10*
SDG 6	**9**	**11**	**13**	**9**	**11**	**11**	**12**
SDG 7	**16**	**17**	**7**	**16**	**17**	**16**	**14**
SDG 8	10	12	9	13	14	8	8
SDG 9	**2**	**2**	**4**	**2**	**2**	**2**	**2**
SDG 10	12	10	11	15	13	14	16
SDG 11	17	9	17	14	15	15	15
SDG 12	*15*	*16*	*12*	*11*	*12*	*13*	*13*
SDG 13	**5**	**6**	**5**	**6**	**4**	**7**	**7**
SDG 14	11	13	16	8	9	10	11
SDG 15	14	15	15	17	16	17	17
SDG 16	6	5	10	4	6	5	3
SDG 17	**3**	**3**	**1**	**3**	**5**	**4**	**5**
Rank Corr (ρ)	1.00	0.89	0.80	0.89	0.93	0.95	0.89

Notes: Each cell reports the within-SDG coverage-gap rank (1 = largest gap, 17 = smallest gap) under that encoder and assignment method. MPNet (768-d) is the canonical general-purpose encoder; MiniLM (384-d) is a smaller general-purpose encoder; SciBERT (768-d) is a scientific-domain encoder. MPNet and SciBERT share dimensionality (768-d), isolating architecture/domain from the dimensionality drop that confounds the MPNet–MiniLM pair (Section 4.6). LR = canonical supervised logistic regression; MLP = 4-layer/384-hidden network; Zero-shot = nearest-centroid on SDG reference centroids, reported for the canonical MPNet encoder only. Coverage gap is document-weighted (Assumption A19) for all methods. Rank Corr (ρ) is the Spearman correlation of each column's SDG coverage-gap ranks against the canonical MPNet-LR column.

Table 7. Per-SDG research–policy gap under distribution-aware metrics versus the canonical centroid gap (Part 1 of 2: full-corpus exact metrics). SWD = sliced Wasserstein; MMD^2 = squared RBF maximum mean discrepancy; Gauss. W_2 = Gaussian 2-Wasserstein; C2ST = classifier two-sample test AUC. The final row reports Spearman ρ between each metric's ranking and the canonical ranking across the 17 SDGs.

SDG	Canonical	SWD	Chamfer	Hausdorff	Gauss. W_2	Grassmann	linear MMD
1	0.447	0.017	0.470	0.640	0.866	3.932	0.319
2	0.278	0.016	0.381	0.508	0.777	3.753	0.255
3	0.470	0.017	0.409	0.552	0.813	3.653	0.297
4	0.262	0.014	0.387	0.494	0.689	3.342	0.205
5	0.312	0.016	0.405	0.536	0.784	3.699	0.263
6	0.280	0.016	0.404	0.520	0.768	3.697	0.259
7	0.418	0.018	0.417	0.552	0.826	3.735	0.321
8	0.422	0.016	0.435	0.564	0.798	3.610	0.273
9	0.291	0.014	0.388	0.528	0.739	3.497	0.200
10	0.296	0.014	0.436	0.555	0.736	3.397	0.213
11	0.464	0.016	0.423	0.550	0.811	3.645	0.309
12	0.434	0.019	0.473	0.650	0.862	3.676	0.331
13	0.481	0.019	0.387	0.519	0.834	3.722	0.375
14	0.368	0.018	0.409	0.541	0.813	3.784	0.303
15	0.221	0.014	0.403	0.515	0.693	3.444	0.189
16	0.331	0.014	0.397	0.519	0.731	3.373	0.214
17	0.216	0.013	0.372	0.459	0.674	3.318	0.192
ρ vs. canonical	–	0.765	0.588	0.654	0.843	0.473	0.870

Table 8. Per-SDG research–policy gap under distribution-aware metrics versus the canonical centroid gap (Part 2 of 2: sampled metrics; continuation of Table 7). EMD = exact 1-Wasserstein; MMD^2 = squared RBF-MMD; Centroid (samp.) = sampled centroid-gap control column.

SDG	Canonical	EMD	Sinkhorn	Energy	MMD^2	C2ST AUC	Centroid (samp.)
1	0.447	0.701	0.555	0.274	0.119	0.997	0.447
2	0.278	0.603	0.522	0.234	0.109	0.997	0.277
3	0.470	0.682	0.561	0.251	0.110	0.996	0.467
4	0.262	0.573	0.507	0.184	0.088	0.997	0.261
5	0.312	0.629	0.531	0.234	0.107	0.996	0.311
6	0.280	0.600	0.536	0.241	0.115	0.998	0.280
7	0.418	0.654	0.545	0.284	0.130	0.998	0.418
8	0.422	0.666	0.534	0.230	0.102	0.995	0.423
9	0.291	0.620	0.523	0.175	0.078	0.997	0.291
10	0.296	0.629	0.591	0.183	0.083	0.991	0.297
11	0.464	0.659	0.530	0.263	0.119	0.997	0.469
12	0.434	0.699	0.600	0.287	0.125	0.998	0.434
13	0.481	0.650	0.489	0.330	0.150	0.997	0.479
14	0.368	0.629	0.546	0.275	0.127	0.996	0.369
15	0.221	0.564	0.521	0.173	0.083	0.998	0.222
16	0.331	0.621	0.486	0.186	0.083	0.997	0.334
17	0.216	0.541	0.446	0.177	0.084	0.995	0.216
ρ vs. canonical	–	0.907	0.439	0.772	0.664	-0.061	0.998

5 Discussion

This section interprets the findings reported above. It moves from what the data show to what they mean: the structure behind the two divergence dimensions, the robust patterns, and the implications and limits of the study. No new results are introduced here.

The headline result of this study is a structural dissociation: the coverage gap — how differently research and policy prioritise each SDG — and the within-SDG semantic gap — how differently they frame the same SDG — are independent dimensions of divergence, not linked effects. The primary test (H1a) finds no detectable association in the canonical specification ($r = -0.000$, $p = 0.999$; $n = 17$). Given the small number of SDG-level observations, this should be interpreted as evidence against a strong monotonic relationship rather than evidence of exact independence. All four hypotheses (H1a–H1d) return null results (Table 2); the null holds across all sensitivity checks excluding SDG 4 or SDG 17. Coverage and framing should therefore be treated as separate axes of divergence, and one cannot be read as a proxy for the other.

This dissociation is possible because the study separates two methodological roles that are often conflated: classification and measurement. The supervised LR classifier (§3.4) is a labeling instrument that assigns texts to SDGs using a transparent 768-coefficient decision boundary. The frozen Sentence-BERT embedding space (§3.3) is a separate measurement surface that provides a consistent geometry for cross-corpus comparison. Classification determines which texts belong to each SDG; semantic distance is then computed independently using centroids in the embedding space. Because the two tasks use different procedures, the dissociation between coverage and framing is a finding about the discourse structure of the corpora, not a design confound. The encoder-sensitivity check independently confirms this dissociation: the coverage ranking is nearly invariant across general-purpose and domain-specialised encoders ($\rho = 0.95$), while the semantic-gap magnitude depends on whether the encoder distinguishes research from policy registers.

The paper makes two contributions, and they carry different weight. The first is a measurement framework that separates classification from measurement, coverage from framing. This framework holds up across changes of assignment method, policy source family, and segment cap, as the checks in Section 4.5 and the appendices show. The second is a set of findings about how AI research and SDG policy actually relate. These findings are weaker: the relationship between coverage and semantic gaps reverses sign under some configurations (Table 2), and SDG 17's position depends heavily on the assignment method (Table 5). The framework should be read as the paper's main contribution. The findings about alignment should be read as a first application, worth checking with a larger and more diverse policy corpus, not as settled results.

The coverage and gap profiles are themselves conditional on corpus composition. The policy source-family split shifts the full-corpus profile: the curated AI/SDG sub-corpus is innovation-heavy (SDG 9 leads), while SDGi and UNGDC are climate-, partnerships-, and governance-oriented (SDGs 13, 17, 16 lead) (Table 18, Appendix H.3). The cross-sensitivity table (Table 3) is therefore the primary interpretive device: it reports gap ranks under three sensitivity axes (policy source, segment cap, and retrieval), and the encoder and assignment-method axes are covered by Table 5 (Section 4.6). The discussion limits its claims to findings that survive this filtering.

5.1 Two Distinct Dimensions of Observed Divergence

This dissociation is best interpreted as a structural difference between knowledge systems rather than as a simple failure of communication. The Mode 1/2 distinction (Gibbons et al., 1994) sharpens this reading. The research corpus concentrates on technically tractable applications in health, energy, and infrastructure — SDGs 3, 7, and 9 — where dominant ML tasks align with well-defined problem-solving rather than institutional coordination, while the policy-side dominance of SDGs 13, 16, and 17 reflects Mode 2 goal-setting arising from contexts of application that demand coordination and collective action. Epistemic-community theory limits the claim: if expert influence partly depends on shared problem framing under uncertainty (Haas, 1992), then large within-SDG semantic gaps mark where such shared framing is less plausible. Following Cross (2013), however, semantic proximity is only a possible precondition for knowledge transfer; it cannot show whether expert communities exist, whether they have policy access, or whether policy actors use the relevant research. The two-communities theory (Caplan, 1979) reinforces this structural reading: the uncorrelatedness of research coverage and semantic gap is what one would expect if the two corpora represent distinct knowledge cultures with different reward systems and knowledge-use criteria, rather than two sides of a shared enterprise.

The practical implication is that observed research–policy divergence is a multidimensional problem: coverage and framing must be measured as separate axes, and neither can serve as a proxy for the other.

5.2 Robust Patterns of Divergence

The cross-sensitivity comparison narrows the set of gap findings stable enough for individual interpretation. SDG 15 is the robustly lowest-gap SDG under the canonical LR and MLP classifiers (rank 16 LR, rank 17 MLP) but sits at rank 12 under zero-shot. SDG 17's position shifts dramatically by assignment method: it is rank 17 (smallest gap) under LR, rank 13 under MLP, but rises to rank 7 under zero-shot nearest-centroid — the most method-sensitive SDG in the study. This confirms that partnership discourse was the most fragile measurement point under the centroid-based approach and benefits from the sharper decision boundaries of supervised classifiers. The sensitivity is consistent with SDG 17's distinctive character as the dedicated means-of-implementation goal, whose policy discourse centres on coordination and resourcing rather than a shared substantive topic (Le Blanc, 2015); a centroid-based method struggles to anchor this qualitatively different semantic boundary, while the supervised LR isolates a tighter region where research and policy texts converge.

SDG 13, SDG 11, and SDG 3 are the most robustly high-gap SDGs, staying in the top six ranks across all three assignment methods (LR, MLP, zero-shot) and across cap and policy-source configurations. These are substantive domains (climate action, sustainable cities, health) where research and policy discourse diverge most sharply. The climate gap (SDG 13) is the most consistent high-gap finding — it ranks first under LR, fourth under MLP, and fifth under zero-shot, while staying in the top six under all policy sub-corpora and segment caps (rank 1 in the canonical and no-cap configurations). SDG 3 is high-gap in the canonical configuration and under every segment cap, but drops to rank 9 under the curated AI/SDG policy family, which frames health through an innovation lens.

SDG 15 is the robustly closest SDG across all three assignment methods, occupying the bottom two ranks under LR and MLP (rank 16, rank 17) and rank 12 under zero-shot. It does not prove substantive alignment, but it shows the instrument can separate a close case from divergent ones. SDG 9, previously the closest gap under the centroid-based approach, now sits in the middle-to-lower band with a gap of 0.291 — research and policy still frame infrastructure in relatively close terms, but the gap is no longer the narrowest.

The main coverage patterns are themselves robust. The full policy corpus is dominated by the governance–climate–partnerships block (SDGs 16, 13, and 17). Appendix H.3 notes that this is source-family sensitive — the curated AI/SDG sub-corpus is more innovation-oriented — but the broader institutional profile holds across the UNGDC and SDGi families. The research corpus, conversely, concentrates on technically tractable SDGs (3, 9, 7, 16, 11).

SDGs 7 and 12 are consistently in the upper half of gap ranks across most configurations, while SDG 11 is the most consistently high-gap goal outside the top three (LR 3, zero-shot 2, MLP 1). SDGs 1, 2, 5, 6, 8, 9, 10, 14, and 16 move by four or more ranks between some configurations (e.g. SDG 8 shifts from rank 6 under LR to rank 15 under zero-shot; SDG 9 shifts from rank 12 under LR to rank 1 under the SDGi-only reference). No individual interpretive claims are made about their precise gap positions.

To make the gap measure concrete, two examples illustrate what high and low semantic distance look like at the term level[1]. SDG 13 (Climate Action), one of the widest gaps

[1] Distinctive terms are TF-IDF-weighted unigrams from deterministic samples of research and policy texts assigned to each SDG by the LR classifier. Full details and additional cases are reported in

(rank 1, gap 0.481), epitomises the high-gap pattern: research texts emphasise *model*, *prediction*, *neural*, and *analysis* — technical measurement language — while policy texts emphasise *climate*, *change*, *emissions*, *countries*, and *action* — institutional commitment language. Both sides concern the same goal, but with fundamentally different orientations: one measures, the other commits. SDG 17 (Partnerships), by contrast, shows the narrowest gap under the supervised classifier (rank 17, gap 0.216): research texts are distinguished by technical terms (*model*, *method*, *machine*) and policy texts by institutional terms (*countries*, *nations*, *global*, *agenda*), yet both sides converge on a shared core of partnership and cooperation vocabulary that keeps the embedding centroids close despite peripheral lexical differences. Together, these cases show that the semantic gap captures a real difference in how the two discourses frame shared SDG assignments, not a uniform property across all goals.

5.3 Implications

Two implications follow from the combined findings, and each argues against reading topical mention as alignment.

Research and policy agenda-setting. The map does not imply that funding should be redirected on the basis of embedding distances; it shows where research–policy translation and brokerage are most needed. In Guston's terms, high-gap SDGs may be candidates for boundary-organising practices that create usable boundary objects while remaining accountable to both knowledge producers and policy users (Guston, 2001).

SDG-aware research communication. Increasing research output on neglected SDGs is insufficient; research communication norms also matter. Future work could test whether publication incentives that encourage explicit SDG framing improve semantic proximity more effectively than topic-level funding mandates alone. In practice, this could mean funder-required policy-facing summaries, journal-level SDG justification fields, or policy synthesis briefs. Following Cash et al. (2003), this is boundary work rather than communication alone, requiring accountability to both scientific and policy principals (Guston, 2001).

These implications speak to distinct audiences. For SDG monitoring bodies, the finding that shared labels do not imply shared framing challenges keyword-based SDG mapping: coverage and framing should be reported as separate dimensions. For research funders, the results suggest that funding allocation alone is unlikely to close the semantic gap without also attending to how research communicates its policy relevance; portfolio-level semantic gap tracking could complement keyword-based counts by measuring whether funded research moves toward or away from policy discourse over successive cycles. For

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Appendix C.1.

journals adopting SDG classification tags, the finding that framing divergence varies by SDG warns against treating SDG tagging as sufficient evidence of policy-aligned research.

5.4 Limitations

The cross-sensitivity comparison (Table 3) bounds which gap findings are stable enough to interpret. The limitations below address scope conditions that this comparison does not resolve.

Scope: semantic proximity, not substantive alignment or policy influence. The study measures semantic proximity in embedding space, not whether research and policy are substantively aligned or whether research is used by policy actors. Embedding proximity reflects shared vocabulary, not shared agenda, and does not measure policy influence, institutional mechanisms, or knowledge use — these would require methods beyond this study's scope.

Distance metric choice. This study uses cosine distance between mean embeddings. A distribution-aware robustness check (Section 4.7) recomputes the per-SDG research–policy gap under Gaussian-2-Wasserstein, exact optimal-transport Earth Mover's Distance, sliced Wasserstein, energy distance, RBF maximum mean discrepancy, and Chamfer distance on the identical point clouds, holding the assignment fixed. The SDG ranking is preserved (Gaussian-2-Wasserstein $\rho = 0.843$, exact EMD $\rho = 0.907$, sliced Wasserstein $\rho = 0.765$, energy $\rho = 0.772$), and SDG 13 is the most divergent SDG under every metric. Centroids therefore discard real distributional shape — the mean Gaussian-2-Wasserstein distance decomposes into a mean term and a shape (covariance) term, and the shape term accounts for a mean share of $\approx 57\%$ — but that shape does not reorder the SDGs. The one metric that does not track the ranking, the classifier two-sample test (C2ST), returns AUC ≈ 1 in all SDGs and is a separation test rather than a ranker. Sampling stability is verified with a replicate seed: the maximum change in any sampled gap is 0.007.

Corpus scope and retrieval boundaries. The policy corpus represents international institutional SDG discourse rather than the universe of AI sustainability policy. Domestic plans, sub-national policies, and non-English documents are not represented; the research corpus is also English-only. Both biases likely understate Global South perspectives. On the research side, the four-term query prioritises precision over recall, a deliberate choice (Section 3.1) that minimises non-AI noise but may miss specialised AI subfields whose practitioners do not use the four umbrella terms in their titles or abstracts. The OECD.AI methodology (OECD, 2024) addresses this through field-of-study membership (the Artificial Intelligence or Machine Learning field of study in MAG, inherited by OpenAlex) rather than keyword search. The corpus is therefore a high-precision subset rather than a comprehensive census of AI-for-sustainability research. Results are conditional on these boundaries. Appendix H.3 makes the source-family sensitivity explicit.

OpenAlex upstream classification. The research corpus is drawn from OpenAlex, whose SDG-topic tagging is itself a downstream classifier trained on external SDG-labelled data; the SDG assignments entering this study are therefore one step from human judgement and inherit that upstream model's bias. Records that OpenAlex does not tag to any SDG are absent from the retrieval universe, so the corpus reflects OpenAlex's classification boundary rather than the full population of AI-for-sustainability research. This is a method-dependent corpus boundary in the sense of Armitage et al. (2020), and coverage proportions should be read as conditional on it.

Corpus asymmetry: AI-for-SDG scope. As noted in Section 3.1, the research corpus is AI $\cap$ SDG while the policy corpus is closer to a union of institutional SDG discourse and AI governance documents. Between-SDG rankings are conditional on this asymmetric construction.

Assignment-method sensitivity of gap estimates. The semantic gap is conditional on how texts are assigned to SDGs. The domain-encoder sensitivity table (Table 5) provides the comparison: within MPNet, the three assignment methods give varied gap rankings (LR vs. MLP $\rho = 0.83$, LR vs. zero-shot $\rho = 0.45$, Table 5). SDG 17 in particular flips from the smallest gap under LR (rank 17) to a mid-ranked gap under zero-shot (rank 7). The LR estimates are the canonical result; the MLP and zero-shot columns are the uncertainty bounds. The raw LR gaps are retained because they best match what the study measures and are most stable across the other sensitivity axes (policy source family and segment cap, Section 4.5). The encoder axis confirms this is not an artefact of the MPNet geometry: the supervised LR ranking is preserved across the two general-purpose encoders (MPNet vs. MiniLM LR $\rho \approx 0.65$), and the same-dimension scientific-domain encoder (SciBERT) compresses the gap magnitude to 0.05 while preserving the coverage structure ($\rho = 0.95$). The zero-shot method is scoped to a single supervised-versus-nearest-centroid comparison under the canonical encoder (Appendix H.2); its moderate gap-rank correlation with the LR baseline ($\rho = 0.45$) is a method-axis effect (nearest-centroid assignment without a trained decision boundary), not an encoder one.

Register effects on the semantic gap. Semantic distance may capture differences in communicative function (see Biber and Conrad (2009) on the register–genre–style distinction). Research abstracts and policy documents are text varieties designed to do different things, so the gap is a measure of observed textual-semantic divergence, not a purified estimate of substantive disagreement. Appendix F reports a register-adjusted sensitivity check. The single-direction (naive) adjustment compresses the gaps modestly (mean change -0.016) and broadly preserves the relative ordering, whereas the stratified iterative variant reorders individual SDGs: SDG 17 flips from the smallest raw gap to the largest adjusted gap. The spread between raw and adjusted estimates is an uncertainty band around the register-sensitive component. Aggressive multi-direction subtraction risks overcorrecting, and no single adjustment cleanly separates register from substantive topic. The raw gap is retained because it best matches what the study measures.

Cross-sectional design. The study observes one period, 2018–2025, against a policy corpus assembled over a similar timeframe; it cannot address whether alignment is improving or deteriorating, or whether specific policy events (e.g., the EU AI Act 2024, IPCC AR6 2022) have shifted research framing.

6 Conclusion

This paper introduces a measurement framework that separates two dimensions of research-policy divergence — coverage and semantic framing — and demonstrates empirically that these dimensions are independent, not linked. The dissociation between coverage gaps and within-SDG semantic gaps holds across three assignment methods, four policy-source families, and two segment-cap configurations: topical attention is not a proxy for framing proximity.

The framework's contribution is portable. It can be applied to any research-policy interface where both sides share a category system and texts can be embedded in a shared semantic space — conservation science, health governance, or technology assessment, for example. The cross-sensitivity structure (Table 3) provides a reusable template for bounding uncertainty in such comparisons: rather than reporting a single gap ranking, the framework makes the stability of each position transparent across the measurement choices that analysts control. The findings about AI-for-sustainability alignment specifically are a first application of this approach, not settled results.

This measurement framework does not solve the alignment problem. It operationalises one dimension of it — semantic proximity in embedding space — and makes the boundary conditions of that operationalisation explicit. Semantic distance cannot tell us whether policy actors use research, whether epistemic communities exist, or whether observed divergence is productive or problematic. What it can do is show, at scale and with verifiable assumptions, where closer investigation is most warranted.

Alignment should therefore be understood as a multi-level empirical construct — from lexical overlap through semantic proximity to substantive agreement (Section 2.2) — rather than as a binary property inferred from shared labels. The framework developed here advances the measurement from Level 1 to Level 2. Level 3 remains the domain of qualitative, institutional, and deliberative methods that this study does not attempt.

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A Supplementary Methodology

A.1 Retrieval and Query Chain

The research corpus was retrieved from the OpenAlex API (Priem et al., 2022) using 68 queries formed by the cross-product of OpenAlex's native SDG work filter and four AI search terms (`artificial intelligence`, `machine learning`, `deep learning`, `neural network`), restricted to `publication_year > 2017`. The 2018 start date reflects the post-AlexNet (Krizhevsky et al., 2012) emergence of deep learning as the dominant paradigm and a research-uptake lag after the SDGs' adoption in 2015. OpenAlex SDG metadata defines the candidate universe; the SDG assignments used in the analysis are produced by the independently validated supervised classifier (Section 3.4). Complete query construction and credential handling are implemented in `1_code/0_fetch/fetch_openalex.py`.

The retrieval boundary is a deliberate scope condition. Elsevier's data-driven characterisation of AI research draws on over 700 field-specific keywords across some 600,000 documents (Hellwig et al., 2019), favouring recall. The OECD.AI Observatory methodology instead defines a paper as AI when it carries the Artificial Intelligence or Machine Learning field of study in the Microsoft Academic Graph (MAG), whose Fields of Study taxonomy OpenAlex later inherited and extended by mapping it to Wikidata identifiers (OECD, 2024; Priem et al., 2022; Wang et al., 2020). The present design prioritises precision: a four-term query keeps non-AI noise out of the embedding space at the cost of recall in specialised AI subfields. Schoeberl et al. (2023) reach the same conclusion for AI-research identification, namely that results are sensitive to the retrieval method and that supervised classification is preferred.

A direct test of this scope condition re-ran retrieval using field-of-study membership for the same two AI fields, mirroring the OECD.AI definition (OECD, 2024), and re-scored the resulting corpus through the identical pipeline. Concept retrieval used OpenAlex concept tags for Artificial Intelligence and Machine Learning, each capped at `MAX_PAPERS_PER_CONCEPT` = 25,000 papers, yielding the roughly 50,000-paper concept corpus. This scale sits on the empirical variance plateau established by the sample-stability ladder (Appendix D): at 50,000 papers the coverage-profile standard deviation is 0.001 and the mean within-SDG semantic gap is 0.354, within 0.002 of the full-corpus estimate of 0.352 (full corpus N = 3,105,144), while policy-text calibration bias has also converged (0.106). The scale also exceeds every corpus size used in prior SDG classification and mapping benchmarks, namely the OSDG community dataset (30,534 texts; Pukelis et al. (2022)), the SDG benchmark encoder evaluation corpus (35,001 papers; Gjorgjevikj et al. (2025)), the SDGi corpus (20,267 segments; SDGi (2024)), the Aurora expert-validated set (4,971 texts; Vanderfeesten et al. (2020)), and the 6,000-paper Kashnitsky et al. benchmark (Kashnitsky et al. (2024)), while remaining far smaller than the full 3.1-million-paper corpus and thus a computationally tractable substrate for OpenAlex concept tagging.

A.2 Segmentation Mechanics

Each source document was segmented using the embedding model's own tokenizer: NLTK sentence-boundary detection (`sent_tokenize`) is followed by greedy accumulation of sentences up to a token budget of `max_seq_length - 10` (margin 10, empirically verified to reduce truncation to near zero). Segments falling below 20 words (`MIN_SEGMENT_WORDS`) are discarded; a single sentence exceeding the token budget is kept whole because mid-sentence splitting is not permissible. For the UN General Debate Corpus, paragraph-level keyword filtering precedes segmentation, so only SDG-relevant speech passages enter the pipeline; this pre-filtering is why UNGDC's segment count (6,128) is much smaller than its source-speech count. OSDG and the SDG Classification Benchmark bypass segmentation because they already consist of short single-label texts. The reference corpora (Knowledge Hub, SDGi-labels, Aurora) and the policy sources are segmented with the same procedure. Segment identifiers encode source document identity for document-level weighting; cross-source deduplication by exact text match (SDGi retained first) yields 40,597 unique policy segments.

A.3 Truncation Fix

During pipeline development, a truncation issue was discovered and corrected: texts exceeding the encoder's maximum sequence length were being silently truncated before embedding, discarding tail content. The fix uses the model's own tokenizer during segmentation (Appendix A.2) to enforce a token budget of `max_seq_length - 10`, verified by the `verify_truncation_rate` utility to produce near-zero post-segmentation truncation. The full corpus was re-embedded after the fix; all reported results use post-fix embeddings.

A.4 Reference-Corpus Provenance

The five reference corpora enter the pipeline at different sizes before preprocessing: OSDG 30,534 raw texts; SDG Classification Benchmark 616; IISD SDG Knowledge Hub 9,172; Aurora 4,022; SDGi 4,225 documents. After deduplication and English filtering, the pooled reference set feeds the single-label filter (Section 3.4), which retains only records with exactly one SDG label, yielding the per-source pool counts reported there (OSDG 30,532; SDGi 20,267; Knowledge Hub 6,122; Aurora 4,971; Benchmark 281) and the 62,173 total. OSDG and the Benchmark bypass segmentation, so their pool counts equal their deduplicated text counts; the other three corpora are segmented before pooling, which is why their pool counts are segment counts.

B Diagnostics for Reference Classifier

B.1 SDG 4 Lexical Artefact Audit

The SDG 4 concern is not used to relabel research records. Its narrower purpose is to test whether the concern is empirically plausible. Table 9 therefore compares the lexical composition of SDG 4-assigned research records with a matched non-SDG4 sample and with SDG 9, using only two transparent keyword groups: ML-learning vocabulary and education-specific vocabulary.

The audit points to a more nuanced conclusion than a pure false-positive story. More than half of the SDG 4-assigned set contains education-specific vocabulary without ML-only terms, and roughly another third contains both education and ML-learning vocabulary. By contrast, the matched non-SDG4 sample and the SDG 9 comparison set are dominated by ML-only vocabulary. This means the SDG 4 signal is not simply the same technical language that drives SDG 9. However, the large “both” share also confirms that ML-learning language and education language overlap materially within the SDG 4-assigned set.

The main analysis therefore retains the canonical LR argmax assignment but continues to treat SDG 4 as artefact-affected. The most accurate reading is a learning-related assignment whose interpretation is blurred by lexical overlap — neither “clean education” nor “pure ML artefact.”

Table 9. SDG 4 lexical artefact audit.

Corpus subset	N	ML-only %	Education-only %	Both %	Neither / unclear %
SDG 4-assigned research	454803	10.3	45.4	23.9	20.5
Non-SDG4 research sample	454803	50.4	4.1	2.8	42.8
SDG 9-assigned research	575703	59.6	2.6	2.9	34.9

Notes: The audit is not used to reassign SDGs. It only checks whether SDG 4-assigned research is disproportionately dominated by machine-learning vocabulary, education vocabulary, or both.

B.2 Centroid Similarity

This subsection reports the pairwise cosine-similarity heatmap (Figure 7) for the 17 canonical SDG centroids. The centroids are the per-SDG mean training embeddings (L2-normalised) used by the LR classifier. The heatmap provides supplementary geometric context for the classification geometry that Section 3.4 builds on.

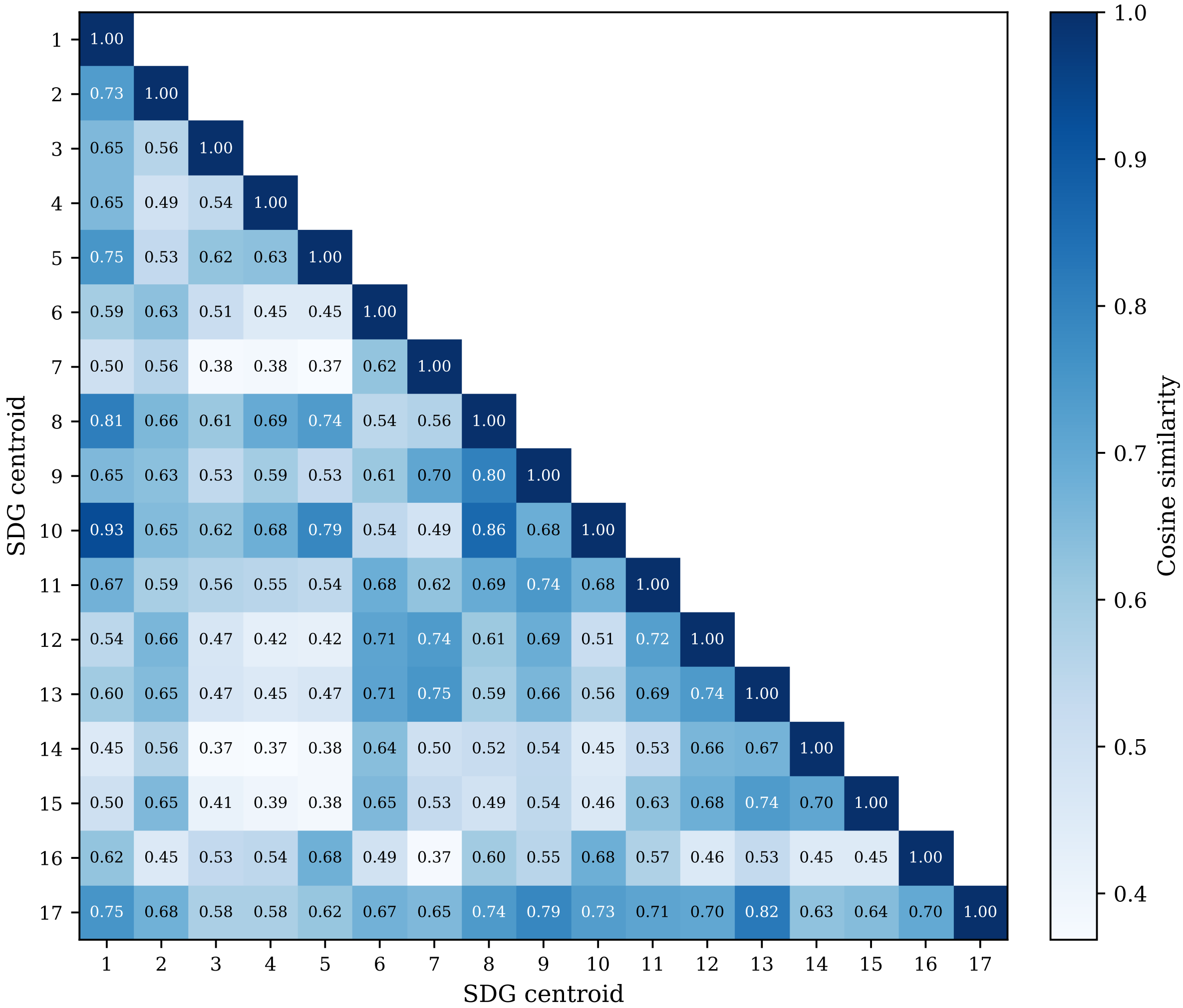

Figure 7. Pairwise cosine similarity between canonical SDG centroids (lower triangle). SDG 13 and SDG 17 sit close together (cosine 0.820), as do SDGs 1, 8, and 10.

Figure 7 shows the full 17×17 similarity grid. The lower triangle is annotated; the upper triangle is left blank because the matrix is symmetric. Reading the grid, three observations stand out. First, the diagonal dominates: most centroids are far more similar to themselves than to any neighbour, which is why the supervised LR classifier (Table 1) can build on this well-separated base to reach 0.817. Second, a small set of off-diagonal cells is elevated. SDG 11's centroid neighbours SDG 9 (cosine 0.745), and SDG 8 lies near SDGs 1 and 10 (cosine 0.806 and 0.863), and SDGs 1 and 10 are themselves the closest off-diagonal pair (cosine 0.930) — these are the proximities that make the SDG 11 and SDG 8 scores in Section 4.1 indicative rather than exact. Third, SDG 17 is the clear outlier: its row and column are the flattest, and its nearest neighbour is SDG 13 (cosine 0.820), reflecting the overlap between climate and partnership discourse noted in Section 4.1.

C Supplementary Robustness and Sensitivity Checks

C.1 Lexical Illustration of the Semantic Gap

Table 10 lists the TF-IDF terms most distinctive of the research and policy sides for each SDG. Generic ML methodology vocabulary—such as *model*, *learning*, and *neural*—is excluded because these terms reflect the search strategy rather than SDG-specific framing.

The contrasts echo the embedding patterns. The widest-gap SDGs (13, 11, 3, 1, 7, 8, 12; $\rho \geq 0.425$) consistently oppose technical measurement vocabulary—*detection*, *images*, *yield*, *regression*—against institutional or welfare terms—*climate*, *emissions*, *health*, *employment*. SDG 13 clarifies the pattern most sharply: research terms centre on crop monitoring and soil science while policy terms address national emissions commitments. Lower-gap SDGs (17, 15, 4, 6, 2, 9; $\rho \leq 0.289$) show narrower lexical divergence, such as the shared technology—infrastructure frame of SDG 9 or the social-media—partnership overlap of SDG 17. A further observation cuts across both groups: agricultural-AI vocabulary (*crop*, *soil*, *agricultural*) appears on the research side of nine unrelated SDGs, reflecting a strong agri-tech skew in the AI-for-SDG literature rather than alignment with each goal's policy domain.

C.2 Implications for Interpreting the Main Estimates

These diagnostics are methodological stress tests that clarify the limits of the LR classifier approach, rather than failed repairs. The PCA diagnostics show that real-world SDG discourse is fuzzy rather than cleanly clustered, especially in the research corpus. The SDG 4 audit shows that the concern is real, but more nuanced than a pure machine-learning false positive. The policy source-family split shows that the full-corpus policy profile is a broader institutional SDG discourse rather than a universal AI policy profile. The canonical hard-threshold classifier assignment (argmax) is retained because it is transparent, reproducible, and directly interpretable.

These analyses also show that the LR classifier should be interpreted as a transparent semantic anchoring procedure rather than a definitive operational SDG system. Attempts to address multi-label overlap and corpus-register bias improve some aspects of the design but introduce new modelling choices. In this paper, the hard-threshold classifier assignment is therefore retained as the canonical specification, while the alternatives are reported as robustness and boundary-condition checks.

Table 10. Distinctive TF-IDF terms for all 17 SDGs.

SDG	Gap	Research-side distinctive terms	Policy-side distinctive terms
1	0.447	poverty alleviation, china, approach, studies, indonesia, literature, zakat, household, problem, satellite	social, cent, national, people, protection, services, population, vulnerable, living, million
2	0.278	crop, detection, diseases, images, plant, soil, precision, leaf, techniques, yield	population, food, children, prevalence, access, worst, hunger, million, rate, births
3	0.470	patients, cancer, detection, disease, clinical, images, cells, potential, genes, diagnosis	health, services, care, national, births, population, mortality, rate, people, access
4	0.262	brain, neurons, cognitive, sleep, cells, food, agricultural, expression, poor, activity	education, school, secondary, primary, national, children, percent, quality, access, vocational
5	0.312	stress, sleep, poor, patients, menstrual, language, understanding, differences, pain, emotion	women, gender, violence, equality, national, rights, girls, percent, government, sexual
6	0.280	irrigation, soil, agricultural, crop, salinity, agriculture, sup, yield, sub, potential	water, sanitation, access, drinking, supply, national, cent, services, population, resources
7	0.418	agricultural, poverty, agriculture, food, approach, battery, drying, parameters, poor, irrigation	electricity, access, renewable, clean, worst best, population, affordable, index worst, cent, national
8	0.422	plant, detection, leaf, image, disease, rice, cnn, agricultural, convolutional, identification	employment, growth, gdp, labour, worst, economic, tourism, government, cent, work
9	0.291	agricultural, agriculture, iot, crop, food, detection, cancer, farmers, farming, cells	infrastructure, innovation, national, government, public, countries, strategy, population, access, intelligence
10	0.296	patients, wealth, income inequality, pain, effects, neurons, studies, literature, evidence, visual	countries, people, rights, worst, national, cent, migration, disabilities, best, population
11	0.464	detection, images, fusion, accident, agricultural, breast, approach, object, recognition, convolutional	housing, urban, city, transport, cities, public, national, waste, plan, green
12	0.434	food, agricultural, image, detection, parameters, potential, approach, soil, compounds, techniques	waste, consumption, national, recycling, production, economy, procurement, public, management, circular
13	0.481	soil, crop, agricultural, yield, moisture, rice, stress, images, detection, sub	climate, change, emissions, national, countries, action, global, energy, nations, united
14	0.368	detection, images, imaging, food, cancer, techniques, hyperspectral, tumor, segmentation, bacterial	marine, biodiversity, fish caught, ocean, rate, sea, worst, coastal, population, embodied
15	0.221	soil, crop, agricultural, plant, spectral, sensing, remote, field, yield, spatial	biodiversity, forest, worst, national, protected, population, area, best, species, freshwater
16	0.331	food, patients, image, poor, pain, cancer, brain, covid-, imaging, studies	nations, international, peace, rights, united, security, law, human, world, global
17	0.216	sentiment, tweets, agricultural, covid-, twitter, yang, agriculture, dan, social, zakat	countries, oecd, international, nations, united, agenda, global, financing, report, cooperation

Notes: Terms are TF-IDF-weighted unigrams and bigrams from deterministic samples of research and policy texts assigned to each SDG.

D Sample-Stability Robustness Check

To test whether the headline findings depend heavily on research-corpus size, I reran the downstream research-side analysis on repeated random subsamples drawn from the pre-embedded, pre-scored OpenAlex corpus. For each of 11 sampled tiers (from 1,000 to 2,000,000 papers), I drew 100 independent samples without replacement, recomputed the research coverage profile, rebuilt sampled research centroids, recalculated the mean within-SDG semantic gap, and re-estimated the policy-text calibration-bias summary. Policy-side quantities were held fixed. The full 3,105,144-paper analysis is shown as a deterministic anchor row rather than as another repeated sample.

Table 11. Sample-stability ladder for the research corpus.

Sample Size	Mean SD of SDG coverage shares	Mean within-SDG semantic gap	Policy-text calibration bias
1,000	0.006	0.422 ± 0.019	0.106 ± 0.008
2,000	0.004	0.390 ± 0.015	0.107 ± 0.006
5,000	0.003	0.367 ± 0.009	0.106 ± 0.004
10,000	0.002	0.361 ± 0.006	0.106 ± 0.003
20,000	0.001	0.357 ± 0.005	0.106 ± 0.002
50,000	0.001	0.354 ± 0.003	0.106 ± 0.001
100,000	0.001	0.353 ± 0.002	0.106 ± 0.001
200,000	0.000	0.353 ± 0.002	0.106 ± 0.001
500,000	0.000	0.353 ± 0.001	0.106 ± 0.000
1,000,000	0.000	0.352 ± 0.001	0.106 ± 0.000
2,000,000	0.000	0.352 ± 0.000	0.106 ± 0.000
Full corpus	–	0.352	0.106

Notes: Mean SD of SDG coverage shares is the unweighted mean of the per-SDG standard deviations in research coverage share across the 100 random draws at each sampled tier. The remaining columns report the mean and draw-to-draw standard deviation of the headline downstream metrics. The full-corpus row is deterministic and therefore has no variance term.

Plain-language guide.[2] The stability ladder shows that the clearest practical plateau appears by roughly 200,000 papers, where the mean semantic gap (∼0.353) and policy-text calibration bias (∼0.106) already stabilise to within 0.001 of their full-corpus values. From 500,000 papers upward, draw-to-draw dispersion is negligible. The full 3,105,144-paper result is retained as the primary estimate because it uses all available data while delivering the tightest values, but the substantive conclusions are not artefacts of sample size choice.

[2] The three metrics are: (1) mean SD of SDG coverage shares — how much the topic balance moves across repeated samples; (2) mean semantic gap — how different research and policy language are within the same SDG; (3) policy–research score gap — how much the scoring instrument favours policy text over research text regardless of content.

The larger corpus also enables per-SDG subgroup detail and tighter rank-order precision that the plateau minimum does not guarantee.

E Model Selection: Grid Search Protocol and Rationale

E.1 Models Tested

Three model families were evaluated under 5-fold GroupKFold cross-validation (document-level grouping, not random splits) on the pooled training data (n=52,835). The test set (n=9,338) was never touched during any grid search.

- **Logistic Regression (LR)**: Interpretable per-dimension coefficients per SDG; fast; standard multiclass baseline.
- **MLP (PyTorch)**: Can capture non-linear interactions in the embedding space.
- **RF / XGB**: Considered but excluded because the embedding space is already a learned representation — bagged trees add ensemble complexity without a clear mechanism for improvement over LR/MLP, and would not produce interpretable coefficients per dimension.

E.2 LR Grid Search

The LR grid searched C $\in \{100, 10, 1, 0.1, 0.01\}$ with L2 penalty. The best configuration was C=10.0, penalty=l2, class_weight=None, solver=lbfgs (macro-F1 = 0.8107 $\pm$ 0.0066 CV). An extended grid adding L1 penalty (l1_ratio=1) and balanced class_weight found no improvement: L1 was 6$\times$ slower and scored 0.8066, while balanced class_weight scored 0.8070. The embedding space is dense — L1 sparsity offers no benefit — and the dual correction of weighted loss plus macro-averaged scoring degrades majority-class performance without compensating minority classes. C=10.0 and C=1.0 are essentially tied (difference $\sim$0.001), but C=10.0 was selected for conservatism.

E.3 MLP Grid Search

The MLP swept n_layers $\in \{1, 2, 4, 8, 16\}$ $\times$ hidden_size $\in \{256, 384\}$ (20 configurations), followed by a learning-rate sweep (lr $\in \{1\mathrm{e}{-4}, 3\mathrm{e}{-4}, 1\mathrm{e}{-3}, 3\mathrm{e}{-3}\}$) on the best architectures. The best MLP (4 layers, 384 hidden, lr=3e-4) achieved macro-F1 = 0.8243 $\pm$ 0.0058. Learning rate had negligible impact — all combinations within $\sim$0.003 of the best.

E.4 Unified Ranking and Final Selection

Rank	Config	CV macro-F1
1	MLP 4L/384h	0.8243 ± 0.0058
2	MLP 4L/256h	0.8242 ± 0.0066
3–15	MLP (2L–4L variants)	0.819–0.824
16–26	MLP (8L–16L variants)	0.817–0.821
27	LR C=10, L2	0.8107 ± 0.0066
28	LR C=1, L2	0.8093 ± 0.0065

Champion: LR (C=10.0, L2). The gap between LR and MLP is 0.0137 macro-F1 ($\sim$1.5 pooled σ), a modest difference insufficient to justify the complexity trade-off. MLP requires 4 hidden layers, AdamW, learning rate tuning, early stopping, and dropout selection (36 configurations swept). LR requires one free parameter (C), no hidden layers, no dropout, no random seed sensitivity. For this study's purpose — building an interpretable supervised reference to compare corpora, not a classifier built to label single documents — LR is the right choice. LR provides 768 signed coefficients per SDG class, each directly attributable to a specific semantic dimension; MLP's hidden layers diffuse this information across non-linear transformations. The macro-F1 numbers are reported honestly: LR 0.811 $\pm$ 0.007, MLP 0.824 $\pm$ 0.006, gap $\sim 1.5\sigma$. LR is chosen for transparency, not superior accuracy.

E.5 Held-Out Test Evaluation of the MLP Robustness Check

The MLP test macro-F1 (0.830) cited in Section 4.1 is a separate evaluation from the CV grid-search results above. After the CV sweep, the champion MLP architecture (4 layers, 384 hidden, lr=3e-4) was retrained on the full training pool (n=52,835) and evaluated once against the held-out test set (n=9,338), the same sanctioned single-use test evaluation applied to the LR champion. The resulting test macro-F1 of 0.830 is not directly comparable to the CV mean of 0.8243 ± 0.0058 from the grid search: the CV number is an average across five document-grouped folds on the training pool alone, while the test number is a single evaluation on a disjoint held-out sample after retraining on the full pool. Both numbers confirm that the LR—MLP gap is modest and that architecture choice is not the binding constraint on classification quality.

F Register-Adjustment Sensitivity

F.1 Naive Single-Regression Approach

This appendix reports an exploratory register-adjustment procedure that probes whether broad research-policy register differences contribute to the raw semantic gap. Following Biber and Conrad's distinction between register, genre, and style (Biber & Conrad, 2009), the research–policy contrast is treated primarily as a register contrast: the two corpora differ in situation of use, audience, institutional setting, and communicative purpose, not merely in surface wording. A learned research–policy direction in embedding space may therefore capture a mixture of modality, abstraction, institutional stance, and substantive theme. The adjusted results are reported as a sensitivity diagnostic, not as corrected estimates.

Procedure. One global research-versus-policy direction is estimated in the shared embedding space using a binary logistic-regression classifier trained on a balanced sample of 79,988 embeddings (approximately 40,000 per class). On the training set, this classifier achieves accuracy 0.983, showing that a strong corpus-level separation axis exists. This degree of separability is expected for two large corpora drawn from distinct institutional discourse communities and does not entail that the within-SDG gaps are primarily register-driven. Let $\mathbf{g}$ denote the learned coefficient direction (L2-normalised). For each embedding $\mathbf{x}$, the orthogonal projection onto $\mathbf{g}$ is subtracted:

$$\mathbf{x}^{\perp} = \mathbf{x} - \frac{\mathbf{x} \cdot \mathbf{g}}{\|\mathbf{g}\|^2}\mathbf{g},$$

and the residual is renormalised to unit length. The result is described as *register-adjusted*, not "debiased": the subtraction removes one estimated global corpus axis, not every possible register or corpus-composition difference.

For the research side, raw per-SDG means are recovered as $\mathbf{c}_j^{\text{res}} = \text{research\_centroid}_j \times \text{research\_cohesion}_j$ (since cohesion equals the norm of the raw mean for L2-normalised embeddings), adjusted, and renormalised. For the policy side, the per-SDG means are computed directly from individual adjusted policy embeddings. Adjusted gaps are then $1 - \cos(\text{adj\_research\_centroid}_j, \text{adj\_policy\_centroid}_j)$.

Results. The high separability between research and policy texts is expected: they represent distinct institutional genres with different audiences, communicative purposes, and stylistic conventions. The logistic-regression classifier's training accuracy (0.983) captures this genre-level distinction, not a measurement of whether within-SDG gaps are real. Importantly, register adjustment reduces rather than eliminates semantic gaps. This suggests that register contributes to observed divergence but does not account for it fully.

On average, the mean gap compresses from 0.352 to 0.336 (a change of -0.016), and the relative ordering of high-gap and low-gap SDGs is broadly preserved: SDGs 13, 3, 1, 11, and 12 remain the furthest apart, while SDGs 15, 17, and 4 remain the closest. The adjustment therefore bounds the register-sensitive component of the gap without replacing the raw estimate.

This single-direction subtraction is a limited diagnostic. It attenuates one broad corpus-level axis but does not cleanly separate register from substantive divergence. The residual gap after adjustment is not a "pure" substantive gap; it only shows what the gap looks like after removing the strongest single linear discriminant between the two corpora. The results are best interpreted as an uncertainty band around the register-sensitive component of the measurement.

Table 12. Register sensitivity check for the semantic gap. Raw gap = the main semantic-gap estimate. Naive = after subtracting one global research–policy direction. Iterative = after removing orthogonal directions until classification accuracy reaches chance.

			Naive		Iterative	
	Description	Raw gap	Adj. gap	Δ	Adj. gap	Δ
13	Climate Action	0.4808	0.4649	-0.0159	0.2346	-0.2462
3	Good Health and Well-Being	0.4698	0.4592	-0.0106	0.2896	-0.1801
11	Sustainable Cities and Communities	0.4645	0.4531	-0.0114	0.2314	-0.2331
1	No Poverty	0.4465	0.4536	0.0070	0.2766	-0.1700
12	Responsible Consumption and Production	0.4342	0.4154	-0.0188	0.2478	-0.1864
8	Decent Work and Economic Growth	0.4224	0.4106	-0.0118	0.2421	-0.1803
7	Affordable and Clean Energy	0.4181	0.3968	-0.0213	0.1693	-0.2487
14	Life Below Water	0.3683	0.3540	-0.0143	0.1317	-0.2365
16	Peace, Justice and Strong Institutions	0.3310	0.3119	-0.0191	0.2665	-0.0645
5	Gender Equality	0.3116	0.3021	-0.0095	0.1835	-0.1281
10	Reduced Inequalities	0.2964	0.2880	-0.0084	0.2572	-0.0392
9	Industry, Innovation and Infrastructure	0.2915	0.2643	-0.0272	0.2412	-0.0503
6	Clean Water and Sanitation	0.2800	0.2561	-0.0239	0.1269	-0.1531
2	Zero Hunger	0.2779	0.2538	-0.0240	0.0937	-0.1842
4	Quality Education	0.2620	0.2275	-0.0345	0.1615	-0.1006
15	Life on Land	0.2208	0.1911	-0.0297	0.0865	-0.1343
17	Partnerships for the Goals	0.2157	0.2104	-0.0053	0.3878	0.1720
Mean		0.3524	0.3360	-0.0164	0.2134	-0.1390

F.2 Stratified Iterative Register Removal

To test whether the single projection exhausts register signal, I iteratively removed separating directions whose logistic-regression classifier is trained on SDG-stratified samples. At each iteration, exactly 1000 research and 1000 policy embeddings are drawn from each SDG, preventing the classifier from learning topic membership as a proxy for register.

Each iteration orthogonalises the new direction against all previous ones, then held-out accuracy is measured on a stratified 15% test split of the projected sample.

Held-out accuracy dropped from 0.981 (iteration 1) to 0.499 by iteration 94, at which point no further linear register separation was detectable. The Spearman rank correlation between the iteration 1 and iteration 94 gap vectors was 0.412: the additional removed directions shifted the relative SDG ordering substantially. The single largest move is SDG 17, which flips from the smallest raw gap (0.2157, rank 17) to the largest adjusted gap (0.3878, rank 1) under iterative removal (Table 12).

Table 13. Stratified iterative register-removal diagnostic. Test acc. = held-out accuracy on a stratified 15% test split after projecting out the first k learned directions. Mean gap = average within-SDG semantic gap after k removals. Spearman ρ = rank correlation of the gap vector at iteration k versus iteration 1.

Iteration	Test acc.	Mean gap	Spearman ρ vs iter 1
1	0.981	0.324	1.000
2	0.938	0.317	0.993
3	0.890	0.305	0.983
4	0.854	0.301	0.953
5	0.833	0.298	0.941
10	0.762	0.263	0.691
15	0.694	0.229	0.515
20	0.619	0.216	0.497
30	0.554	0.207	0.449
40	0.541	0.205	0.461
50	0.522	0.205	0.456
94	0.499	0.213	0.412

The identification argument rests on the stratification design. Because every iteration trains on equal numbers of research and policy embeddings from each SDG, the classifier cannot learn which SDG a document belongs to — every SDG is balanced across classes. The only remaining signal available for discrimination is the corpus-level register difference: the way research papers sound versus the way policy documents sound, within the same substantive topic. What gets removed at each iteration is therefore precisely the register component — modality differences (academic prose versus policy prose), audience conventions (peer reviewers versus policymakers), institutional stance (hedged claims versus prescriptive language), and structural patterns (methods sections versus action items). What survives is the semantic content difference between AI research and SDG policy for each specific SDG.

The results show that register is real but finite. The mean gap compresses from 0.324 to 0.213, a 34% reduction, before plateauing. The Spearman ρ decays rapidly in the first 15 iterations (1.000 to 0.515) and then stabilises around 0.455, indicating that the first 15 orthogonal directions capture the bulk of the register signal. The plateau is the key finding: it demonstrates that the semantic gap measured in the main text is not an artefact of corpus-level register differences. Even after removing all linear register signal (classification accuracy reaching chance at 0.499), the within-SDG gaps persist and the mean gap stabilises at a non-zero value. The SDG ranking is not preserved, however: the correlation between the iteration-1 and final gap vectors is only 0.412, and SDG 17 in particular flips from the smallest raw gap to the largest adjusted gap (Table 12). The adjusted estimates therefore bound the register-sensitive component of the gap, and the main text retains the raw estimate as the canonical measure.

G Concept-Retrieval Sensitivity

Advisor review flagged the four-term keyword query used to retrieve the AI research corpus as an intuitive choice. To test sensitivity to that choice, retrieval was repeated using field-of-study membership for the two AI fields of study (OECD, 2024), mirroring the OECD.AI Observatory methodology, which treats a paper as AI if tagged with the Artificial Intelligence or Machine Learning field of study in the Microsoft Academic Graph (MAG), the taxonomy that OpenAlex later inherited and extended (Priem et al., 2022; Wang et al., 2020). No SDG filter was applied. SDG assignment was performed by the same trained logistic-regression classifier used throughout (Section 3.4), so the only difference from the main corpus is retrieval strategy. The resulting 50,000-paper corpus was passed through the identical segmentation, embedding, scoring, and analysis pipeline.

Table 14 compares the SDG coverage shares of the keyword-retrieved and concept-retrieved corpora. The two profiles are closely aligned (Spearman $\rho = 0.937$, Kendall $\tau = 0.853$); the largest deviations are SDG 4 (-7.6pp, the known ML-vocabulary artefact, Appendix B.1) and SDG 9 (+6.4pp). The within-SDG coverage-gap and semantic-gap rankings are also stable against the keyword baseline (coverage-gap rank $\rho = 0.88$; semantic-gap rank $\rho = 0.88$, Table 3). Schoeberl et al. (2023) independently find that AI-research identification is sensitive to the retrieval method and recommend supervised classification. The present LR scoring is that recommended step applied to SDG assignment. We therefore report the main findings as robust to the query-term choice, while noting that both corpora remain conditional on their respective retrieval strategies.

Table 14. SDG coverage-share comparison: keyword-retrieved versus concept-retrieved (AI/ML field-of-study) research corpus. Δ is concept minus keyword share in percentage points.

SDG	Keyword %	Concept %	Δ
SDG 1	0.6	0.6	+0.0
SDG 2	2.5	1.9	-0.6
SDG 3	26.0	26.9	+0.8
SDG 4	14.6	6.9	-7.7
SDG 5	2.4	1.6	-0.8
SDG 6	1.1	1.0	-0.1
SDG 7	3.7	4.0	+0.4
SDG 8	2.4	2.7	+0.4
SDG 9	18.5	25.2	+6.6
SDG 10	1.2	0.8	-0.4
SDG 11	4.9	6.8	+1.9
SDG 12	2.2	3.9	+1.7
SDG 13	4.2	5.4	+1.2
SDG 14	1.9	1.5	-0.3
SDG 15	2.4	2.4	+0.0
SDG 16	10.9	7.8	-3.1
SDG 17	0.5	0.5	+0.0
Spearman $\rho = 0.937$; Kendall $\tau = 0.853$			

H Supplementary Cross-Method Data

H.1 Cross-Method Coverage and Semantic Gap Values

Tables 15 and 16 report the raw coverage gap and semantic gap values for each SDG across seven encoder–assignment-method combinations, with an additional concept-retrieval LR column for comparison against the keyword-retrieved research corpus baseline (Appendix G). While Tables 3 and 4 rank these values to compare cross-method stability, the tables below show the unrounded gap magnitudes, allowing the rank separation to be interpreted in absolute terms. Coverage gap is document-weighted (Assumption A19); semantic gap policy centroids use a 50-segment-per-document cap. The zero-shot nearest-centroid method is reported for the canonical MPNet encoder only; it is scoped to a single supervised-versus-nearest-centroid comparison (Appendix H.2), not a cross-encoder axis.

H.2 Assignment-Method Comparison: Supervised vs. Nearest-Centroid

The zero-shot nearest-centroid method is used in this study for one purpose only: a single comparison between the canonical supervised assignment (LR, with the MLP as robustness) and an assignment rule with no trained decision boundary. Under the zero-shot rule, every

Table 15. Coverage gap values across embedding architectures, assignment methods, and retrieval strategies.

SDG	Coverage gap (%)							
	MPNet			MiniLM		SciBERT		Concept
	LR	MLP	ZS	LR	MLP	LR	MLP	
1	4.2	4.0	3.8	3.8	4.0	4.9	6.4	4.1
2	1.7	2.0	0.7	1.7	1.9	2.4	3.0	2.3
3	21.3	21.3	18.8	23.2	22.0	23.7	24.6	22.2
4	9.9	7.8	11.1	8.7	9.6	12.6	10.9	2.2
5	2.8	3.7	2.6	1.6	2.4	3.3	2.9	3.6
6	2.6	2.3	0.9	2.1	1.8	2.5	2.4	2.7
7	0.6	0.1	3.4	0.3	0.1	1.3	1.7	0.2
8	2.0	2.3	2.0	1.5	1.0	3.5	3.5	1.7
9	14.2	13.0	8.5	10.8	10.5	14.5	12.6	20.9
10	1.7	2.4	1.3	0.7	1.4	1.5	0.2	2.1
11	0.3	3.6	0.4	1.4	0.6	1.5	1.0	1.6
12	1.4	1.0	1.1	1.7	1.6	2.3	2.0	0.3
13	7.1	6.2	8.4	5.6	9.5	4.5	3.6	5.9
14	1.9	2.1	0.5	2.2	2.1	2.7	2.6	2.2
15	1.7	1.9	0.5	0.2	0.6	0.6	0.1	1.7
16	6.6	7.1	1.5	10.4	7.5	8.9	12.1	9.7
17	11.0	10.7	20.2	10.5	7.8	10.9	8.5	11.0

Notes: Coverage gap = |research proportion − policy proportion| per SDG in percentage points, using document-weighted policy proportions (Assumption A19) for all methods.

research paper and policy segment is assigned to the SDG whose train-split reference centroid is most similar (cosine) in embedding space. Table 17 reports how often the two methods agree, per SDG and overall, for the research corpus (paper level) and the policy corpus (segment level), together with the LR-versus-zero-shot semantic-gap rank delta per SDG.

The comparison confirms that the two methods agree on the majority of assignments: the overall LR-versus-zero-shot agreement rate is 62.3% over the 3.1M research papers and 79.7% over policy segments (80.4% at document level; Table 17). Agreement is not uniform across SDGs, and the disagreements are concentrated where the reference centroids are least separated, in line with the centroid-similarity diagnostics (Appendix B.2). SDG 17, whose reference centroid is the flattest and most isolated in that analysis, is also the most method-sensitive goal here: it shows the largest rank delta between the two semantic-gap

Table 16. (continued) Semantic gap values across embedding architectures, assignment methods, and retrieval strategies.

SDG	Semantic gap (cosine)							
	MPNet			MiniLM		SciBERT		Concept
	LR	MLP	ZS	LR	MLP	LR	MLP	
1	0.447	0.434	0.384	0.378	0.454	0.046	0.064	0.767
2	0.278	0.296	0.328	0.305	0.318	0.042	0.038	0.397
3	0.470	0.429	0.442	0.476	0.474	0.055	0.052	0.600
4	0.262	0.264	0.278	0.288	0.290	0.044	0.043	0.349
5	0.312	0.255	0.247	0.481	0.324	0.051	0.054	0.440
6	0.280	0.321	0.417	0.297	0.322	0.043	0.099	0.344
7	0.418	0.458	0.512	0.404	0.398	0.050	0.049	0.500
8	0.422	0.406	0.265	0.477	0.487	0.035	0.035	0.594
9	0.291	0.310	0.239	0.325	0.375	0.042	0.041	0.384
10	0.296	0.259	0.294	0.465	0.471	0.055	0.047	0.436
11	0.464	0.599	0.462	0.360	0.415	0.049	0.053	0.679
12	0.434	0.428	0.356	0.330	0.401	0.040	0.049	0.414
13	0.481	0.429	0.428	0.499	0.452	0.051	0.053	0.570
14	0.368	0.340	0.453	0.302	0.307	0.044	0.042	0.506
15	0.221	0.242	0.315	0.337	0.299	0.045	0.079	0.296
16	0.331	0.319	0.327	0.304	0.351	0.041	0.040	0.506
17	0.216	0.285	0.399	0.246	0.360	0.037	0.034	0.345

Notes: Semantic gap $= 1 - \cos(\mathbf{r}_{\mathrm{SDG}}, \mathbf{p}_{\mathrm{SDG}})$ where $\mathbf{r}_{\mathrm{SDG}}$ and $\mathbf{p}_{\mathrm{SDG}}$ are the research and policy centroids; policy centroids use a 50-segment-per-document cap. All values are cosine units.

rankings ($|\Delta| = 10$) and one of the lowest per-SDG assignment agreement rates on the research side (16.8%, second only to SDG 10 at 15.9%). SDG 8, whose supervised rank is also heavily method-dependent (rank 6 under LR, rank 15 under zero-shot), shows the second-largest rank delta ($|\Delta| = 9$). The zero-shot rule, having no supervised decision boundary, anchors partnership and inequality discourse less sharply than the LR classifier. The MLP robustness check agrees with the zero-shot rule at 76.8% of policy segments (78.1% at document level), close to the LR-versus-zero-shot rate; research-level MLP agreement is not reported because MLP per-paper scores are not persisted, and the research LR-versus-MLP gap-rank correlation (Section 4.6) proxies it.

H.3 Detailed Policy Profiles by Source Family

The full policy corpus combines a curated AI/SDG sub-corpus, SDGi VNR/VLR reports, and UN General Debate speeches. These are all policy-facing texts, but they are not interchangeable genres. The family split reported in Table 18 therefore asks a narrow

Table 17. Assignment agreement between the supervised LR classifier and the zero-shot nearest-centroid rule, by SDG and overall, under the canonical MPNet encoder and keyword retrieval. Agree % is the share of LR-assigned texts that the zero-shot rule also assigns to the same SDG. Semantic-gap ranks are 1 = largest gap; $|\Delta|$ is the absolute rank difference between the two methods.

	Research (papers)		Policy (segments)		Semantic-gap rank		
SDG	n_{LR}	Agree %	n_{LR}	Agree %	LR	ZS	$\lvert\Delta\rvert$
SDG 1 (No Poverty)	18,270	20.1	1,798	65.4	4	8	4
SDG 2 (Zero Hunger)	78,635	64.6	1,712	93.3	14	10	4
SDG 3 (Good Health)	807,587	74.7	2,926	87.4	2	4	2
SDG 4 (Quality Education)	454,803	80.7	2,063	89.7	15	14	1
SDG 5 (Gender Equality)	73,507	46.3	2,047	87.5	10	16	6
SDG 6 (Clean Water)	34,114	76.5	1,207	94.1	13	6	7
SDG 7 (Clean Energy)	113,520	84.2	1,770	85.1	7	1	6
SDG 8 (Decent Work)	73,410	34.1	1,998	70.2	6	15	9
SDG 9 (Industry & Infra.)	575,703	46.9	4,135	76.7	12	17	5
SDG 10 (Reduced Inequalities)	38,447	15.9	1,212	53.7	11	13	2
SDG 11 (Sustainable Cities)	152,483	69.0	1,631	86.8	3	2	1
SDG 12 (Responsible Cons.)	69,156	41.9	1,223	87.0	5	9	4
SDG 13 (Climate Action)	130,236	36.0	3,394	83.3	1	5	4
SDG 14 (Life Below Water)	57,543	78.0	1,720	87.0	8	3	5
SDG 15 (Life on Land)	74,431	74.9	1,810	83.9	16	12	4
SDG 16 (Peace & Justice)	338,664	49.6	5,408	64.9	9	11	2
SDG 17 (Partnerships)	14,635	16.8	4,543	81.0	17	7	10
Overall	3,105,144	62.3	40,597	79.7			

Notes: Assignment agreement is computed over all texts (research: each paper; policy: each segment), without the 50-segment-per-document cap, which applies only to centroid building. The zero-shot rule assigns each text to the SDG with the highest cosine similarity to the train-split reference centroids ($\mathbf{s}_k$); the LR rule uses the trained classifier's argmax. MLP agreement against zero-shot is computed for the policy corpus only, because MLP research per-paper scores are not persisted (the research LR-vs-MLP gap-rank correlation is reported in Section 4.6).

question: whether the full-corpus policy profile is broadly stable across source families or whether it is strongly shaped by corpus composition.

The answer is mixed. The broad international-institutional profile is clearly source-family sensitive. The curated AI/SDG family is much more innovation-oriented, with SDG 9 leading its document-weighted profile, while the SDGi and UNGDC families are more strongly concentrated on climate, partnerships, governance, and, in the SDGi case, SDG 11-style institutional review language. At the same time, the broader full-corpus pattern is not driven by one family alone: both the UNGDC speeches and the full corpus are dominated by the climate–partnerships–governance block, and SDG 17 remains the largest semantic-gap case across all three families.

The implication is interpretive rather than corrective. The full policy corpus should be read as international institutional SDG discourse, not as a universal profile of all AI sustainability policy. The scope of what the policy-side distribution represents is accordingly narrower, but the main diagnostic purpose remains valid.

Table 18. Per-SDG policy source-family sensitivity: coverage share and semantic gap.

SDG	Full Corpus		Curated AI/SDG		SDGi VNR/VLR		UNGDC	
	pol.% (n)	sem. (n)	pol.% (n)	sem. (n)	pol.% (n)	sem. (n)	pol.% (n)	sem. (n)
1	4.7 (302)	0.45 (1,761)	1.1 (1)	0.46 (387)	6.0 (255)	0.45 (1,210)	2.2 (46)	0.58 (164)
2	4.3 (271)	0.28 (1,494)	0.0 (0)	0.32 (238)	5.8 (245)	0.28 (1,188)	1.3 (26)	0.43 (68)
3	4.7 (299)	0.47 (2,424)	6.4 (6)	0.35 (649)	6.7 (283)	0.53 (1,712)	0.5 (10)	0.69 (63)
4	4.7 (301)	0.26 (2,063)	0.0 (0)	0.21 (288)	6.9 (291)	0.28 (1,704)	0.5 (10)	0.40 (71)
5	5.1 (327)	0.31 (2,005)	0.0 (0)	0.30 (277)	6.6 (277)	0.32 (1,523)	2.4 (50)	0.45 (205)
6	3.7 (238)	0.28 (1,207)	0.0 (0)	0.33 (140)	5.4 (227)	0.28 (1,036)	0.5 (11)	0.42 (31)
7	4.2 (268)	0.42 (1,572)	0.0 (0)	0.50 (366)	5.9 (251)	0.40 (1,138)	0.8 (17)	0.58 (68)
8	4.4 (280)	0.42 (1,905)	0.0 (0)	0.41 (329)	6.5 (276)	0.44 (1,558)	0.2 (4)	0.57 (18)
9	4.3 (275)	0.29 (2,992)	54.3 (51)	0.25 (1,865)	5.1 (214)	0.54 (1,100)	0.5 (10)	0.42 (27)
10	3.0 (189)	0.30 (1,148)	0.0 (0)	0.37 (286)	4.3 (183)	0.30 (797)	0.3 (6)	0.42 (65)
11	5.2 (330)	0.46 (1,631)	0.0 (0)	0.45 (167)	7.8 (328)	0.47 (1,454)	0.1 (2)	0.58 (10)
12	3.6 (229)	0.43 (1,115)	0.0 (0)	0.49 (172)	5.4 (227)	0.44 (937)	0.1 (2)	0.61 (6)
13	11.3 (719)	0.48 (3,096)	7.4 (7)	0.43 (609)	6.0 (252)	0.45 (1,276)	22.5 (460)	0.58 (1,211)
14	3.7 (237)	0.37 (1,296)	1.1 (1)	0.37 (226)	4.5 (189)	0.37 (859)	2.3 (47)	0.50 (211)
15	4.1 (261)	0.22 (1,467)	0.0 (0)	0.27 (291)	5.6 (236)	0.22 (1,097)	1.2 (25)	0.40 (79)
16	17.5 (1,113)	0.33 (4,775)	8.5 (8)	0.21 (962)	6.0 (253)	0.36 (1,402)	41.6 (852)	0.49 (2,411)
17	11.4 (728)	0.22 (3,687)	21.3 (20)	0.21 (912)	5.6 (238)	0.20 (1,355)	22.9 (470)	0.31 (1,420)
ρ	1.00	1.00	0.34	0.76	0.68	0.82	0.48	0.96

Notes: For each policy source family, `pol.% (n)` is the document-weighted policy share (%) with the document count in parentheses; `sem. (n)` is the within-SDG research–policy semantic gap with the capped segment count in parentheses. Read across a row to see whether the gap is stable or driven by one family.

I Declaration of AI Use

In accordance with the University's guidance on generative AI ("Use of GenAI is permitted to support assignments"), this appendix declares the use of AI tools throughout the dissertation. All use was conducted with human oversight and control; AI-generated outputs were reviewed, modified, and independently verified before inclusion. No AI-generated text was submitted without such verification. This declaration follows the order of permitted activities specified in the assessment guidelines. AI-assisted activities not explicitly listed (e.g., code debugging, plotting, LaTeX troubleshooting, literature summarisation) fall under the same oversight and verification standards described in the relevant rows above.

Table 19. Declaration of AI use by permitted activity.

Activity	**AI tool(s)**	**How output was used**
Researching and refining ideas	ChatGPT, MiniMax / DeepSeek (via OpenCode), GPT (via Codex)	ChatGPT was used iteratively to develop ideas that aligned with the author's intent. Coding agents independently checked data and code availability to confirm feasibility. All final decisions and conclusions are the author's own. Final research design was formally presented to and approved by supervisor.
Information retrieval / background research	Gemini, MiniMax / DeepSeek (via OpenCode), GPT (via Codex)	AI tools read the manuscript for context and recommended papers to investigate further. All cited sources were independently verified against the original publications.
Drafting an outline / organising thoughts	MiniMax / DeepSeek (via OpenCode), GPT (via Codex)	Generated code scaffolding and assisted with the full analysis pipeline (embedding, scoring, validation, robustness, appendix outputs). The centroid-assignment step was verified against hand-labeled samples, and the scoring and coverage-gap functions were unit-tested. Analytical decisions were directed by the author.
Refining research questions	ChatGPT, MiniMax / DeepSeek (via OpenCode), GPT (via Codex)	Overlaps with row 1. ChatGPT sharpened framing based on reviewed literature; coding agents confirmed questions were answerable with available data and models. The author retained full control over the final formulation. RQs presented to and approved by supervisor.
Checking spelling and grammar	ChatGPT, Gemini, Claude (web/mobile), MiniMax / DeepSeek (via OpenCode), GPT (via Codex)	Coding agents provided inline stylistic and phrasing suggestions during drafting. Web and mobile versions of ChatGPT, Gemini, and Claude were used for editorial review and proofreading. All suggestions were accepted or rejected per standard editorial judgment.